

TSCG: The Transdisciplinary System Construction Game

- 1 TSCG: Transdisciplinary System Construction Game
 - 1.1 A Systemic Modeling Toolkit
 - 1.2 Changelog
 - 1.2.1 v6.0 — June 2026
 - 1.3 Abstract
 - 1.4 1. Introduction
 - 1.4.1 1.1 The Problem of Disciplinary Silos and the “Desiloification” Hypothesis
 - 1.4.2 1.2 Origin: Twenty-Five Years of Intuition, Then AI-Assisted Concretization
 - 1.4.3 1.3 The Bicephalous Architecture: Stereoscopic Knowledge
 - 1.5 2. The TSCG Architecture
 - 1.5.1 2.1 M3 — The GenesisGrammar: Three Monoidal Grammars
 - 1.5.2 2.2 Gt — The Eagle Eye (Territory / ASFID)
 - 1.5.3 2.3 Gm — The Sphinx Eye (Map / REVOI)
 - 1.5.4 2.4 Gs — The Stereopic Grammar (TKSL)
 - 1.5.5 2.5 M2 — GenericConcepts: The Transdisciplinary Vocabulary
 - 1.5.6 2.6 M1 — Domain Extensions
 - 1.5.7 2.7 M0 — Instances
 - 1.5.8 2.8 The Purity Principle
 - 1.6 3. The GenericConcepts Layer (M2)
 - 1.6.1 3.1 Structural Grammar as a Concept Construction Language
 - 1.6.2 3.2 The Nine Families
 - 1.6.3 3.3 Bicephalous GenericConcepts
 - 1.6.4 3.4 GenericConceptCombos
 - 1.6.5 3.5 The Causal Chain for Irreversibility
 - 1.6.6 3.6 M2 Purity and the Six-Domain Threshold
 - 1.6.7 3.7 Mathematical Validation Status
 - 1.7 4. The Instance Corpus
 - 1.7.1 4.1 Instances as the Empirical Backbone of TSCG
 - 1.7.2 4.2 Poclets (25 validated)
 - 1.7.3 4.3 SystemicFrameworks (3 validated)
 - 1.7.4 4.4 SymbolicSystemGrammars (2 validated)
 - 1.7.5 4.5 TscgTools (3 validated)
 - 1.7.6 4.6 The Simulation Gallery
 - 1.7.7 4.7 SHACL Validation Infrastructure
 - 1.8 5. Epistemic Scoring
 - 1.8.1 5.1 Overview: Two Complementary Metrics
 - 1.8.2 5.2 δ_1 — The Epistemic Gap
 - 1.8.3 5.3 δ_2 / EFS — The Epistemic Focal Score
 - 1.8.4 5.4 The scoringStatus Property
 - 1.8.5 5.5 Scoring as a Research Instrument
 - 1.9 6. Validation
 - 1.9.1 6.1 Epistemological Stance: Falsificationism, Not Proof
 - 1.9.2 6.2 The Six-Domain Threshold
 - 1.9.3 6.3 The Corpus as Empirical Record
 - 1.9.4 6.4 The TriskeleToolchain Validation
 - 1.9.5 6.5 Ontological Overfitting: The Central Anti-Pattern
 - 1.9.6 6.6 Limitations and Open Questions
 - 1.10 7. Discussion
 - 1.10.1 7.1 What TSCG Is — and Is Not
 - 1.10.2 7.2 The Structural Grammar Migration: A Foundational Step
 - 1.10.3 7.3 Two Named Failure Modes: Narcissus and Icarus
 - 1.10.4 7.4 TSCG and Prior Art: Similarities and Distinctions
 - 1.10.5 7.5 The Human-AI Collaboration as a Research Method
 - 1.10.6 7.6 On Being a Work in Gestation
 - 1.11 8. References
 - 1.11.1 Foundational Theory
 - 1.11.2 Systemic Frameworks
 - 1.11.3 Ontology Engineering & Semantic Web
 - 1.11.4 Mathematical & Algebraic Foundations
 - 1.11.5 Human-AI Collaboration
 - 1.11.6 TSCG Primary Sources
 - 1.12 Appendix A — Complete Instance Corpus (v16.2.0)
 - 1.12.1 A.1 Poclets (25)
 - 1.12.2 A.2 SystemicFrameworks (3)
 - 1.12.3 A.3 SymbolicSystemGrammars (2)
 - 1.12.4 A.4 TscgTools (3)
 - 1.13 Appendix B — Human-AI Collaboration Methodology

- 1.13.1 B.1 Infrastructure
- 1.13.2 B.2 Active Skills
- 1.13.3 B.3 Task Delegation Matrix
- 1.14 Appendix C — M2 GenericConcepts Catalog (v16.14.0)
 - 1.14.1 C.1 Structural Family (23 concepts)
 - 1.14.2 C.2 Dynamic Family (17 concepts)
 - 1.14.3 C.3 Ontological Family (10 concepts)
 - 1.14.4 C.4 Regulatory Family (10 concepts)
 - 1.14.5 C.5 Informational Family (8 concepts)
 - 1.14.6 C.6 Adaptive Family (4 concepts)
 - 1.14.7 C.7 Energetic Family (4 concepts)
 - 1.14.8 C.8 Relational Family (5 concepts)
 - 1.14.9 C.9 Teleonomic Family (1 concept)
 - 1.14.10 C.10 Combo Family (4 validated GenericConceptCombos)
- 1.15 Appendix D — Scoring Reference
 - 1.15.1 D.1 ASFID/REVOI Score Properties
 - 1.15.2 D.2 Derived Metrics
 - 1.15.3 D.3 SpectralClass Enumeration
 - 1.15.4 D.4 EFS — Epistemic Focal Score (δ_2)
 - 1.15.5 D.5 FocalClass Enumeration
 - 1.15.6 D.6 ScoringStatus Enumeration
 - 1.15.7 D.7 Reference Implementation
- 1.16 Appendix E — Framework Maturity Assessment (v16.2.0)
- 1.17 Appendix F — Core Hypotheses & Theoretical Foundations
 - 1.17.1 F.1 Core Hypotheses
 - 1.17.2 F.2 Anti-Patterns & Safeguards
 - 1.17.3 F.3 Position Statements
 - 1.17.4 F.4 Extended Theoretical Foundations

1 TSCG: Transdisciplinary System Construction Game

1.1 A Systemic Modeling Toolkit

Authors: Michel Kern (aka Echopraxium) with the collaboration of Claude AI **Date:** June 2026 **Version:** 6.0 **Smart Prompt Version:** TSCG v16.2.0 (June 2026) **DOI (Prior Work):** 10.5281/zenodo.19544443 **Repository:** <https://github.com/Echopraxium/tscg> **Live Demo:** <https://echopraxium.github.io/tscg/> **License:** CC BY 4.0 (document) — BSD 3-Clause Clear (source code)

1.2 Changelog

1.2.1 v6.0 — June 2026

- Complete rewrite aligned with TSCG framework v16.2.0
- Mathematical foundation migrated: Hilbert/tensor → Structural Grammar (Lambek calculus, free commutative monoidal categories)
- M3: Gs/TKSL (Stereopsis grammar) fully documented as autonomous section
- M2: 9 families, 80 GenericConcepts, formulas corrected to $\times / + / |$ operators
- Corpus: 33 validated instances (25 Poclets, 3 SystemicFrameworks, 2 SymbolicSystemGrammars, 3 TscgTools)
- Scoring: δ_1 SpectralClass thresholds corrected; δ_2 /EFS (Epistemic Focal Score) fully documented
- Acronym corrected: SFE → EFS (Epistemic Focal Score)
- Author: “Michel Kern (aka Echopraxium) with the collaboration of Claude AI”
- TriskeleToolchain added as SymbolicSystemGrammar — Base16 ISA validation

1.3 Abstract

TSCG is an **exploratory modeling toolkit** for complex systems, arising from more than twenty-five years of personal intuition about recurrent transdisciplinary invariants. Rather than a validated theory, it is an **open invitation** — an arbitrary but operational modeling grid built around a **bicephalous architecture** combining two complementary perspectives: **ASFID** (Attractor, Structure, Flow, Information, Dynamics) for Territory measurement, and **REVOI** (Representability, Evolvability, Verifiability, Observability, Interoperability) for Map construction. A third grammar, **TKSL** (Temporality, Knowledge, Symbol, Localizability), formalizes their stereopsis fusion — the epistemic depth produced by triangulating the two perspectives.

The mathematical foundation is **Structural Grammar** based on Lambek calculus and free commutative monoidal categories. The monoidal product operators ($\times$, $+$, $|$) capture the simultaneous, non-separable co-presence of dimensions — without requiring a metric space or Hilbert structure — placing TSCG on a rigorous algebraic footing, pending expert review.

The toolkit adopts a **falsificationist stance**: it has not yet been invalidated by a corpus of **33 validated instances** — the primary empirical instrument of TSCG. Historically, the first instance type to emerge was the **Poclet** (25 validated): a minimal, pedagogical model of a single system, spanning photography, Norse mythology, nuclear engineering, biology, electronics, music theory, blockchain consensus, plate tectonics, and the periodic table. The corpus subsequently extended to three additional instance types: **SystemicFrameworks** (VSM, TRIZ, Business Model Canvas), **SymbolicSystemGrammars** (I-Ching, TriskeleToolchain), and **TscgTools** — software instruments operating reflexively on the toolkit itself.

Epistemic alignment is measured by two complementary metrics. The **epistemic gap δ_1** ($|ASFID_mean - REVOI_mean| / \sqrt{2}$) quantifies the distance between Territory measurement and Map construction, classified into four SpectralClasses (Coherent, OnCriticalLine, Liminal, Enigmatic). The **Epistemic Focal Score (EFS / δ_2)** ($\text{stereopsisDepth} \times (1 - |focalBias|)$) measures the quality of stereopsis fusion when Gs/TKSL primitives are mobilized, classified into six FocalClasses from Emmetropic to Astigmatic.

Twelve Poclets are accompanied by standalone HTML simulations (BabylonJS 3D or p5.js Canvas2D) forming the **TSCG Simulation Gallery**. The Business Model Canvas SystemicFramework features the most advanced simulation: 12 real-world company cases, 5 lifecycle phases, and 34 documented transitions.

TSCG is submitted to the research community not as a Theory of Everything, but as a **community-revisable construction kit** — a work in active gestation, submitted not imposed. It is an invitation to test whether this arbitrary grid can be broken, and if not, what it enables us to see across disciplinary silos.

Without close collaboration with Claude AI — including documentary research, delegation of algebraic formalization, ontology encoding, and pair modeling for candidate GenericConcepts — this project would not have been concretized.

Keywords: Systems Modeling, Ontology Engineering, Transdisciplinarity, Knowledge Representation, Map-Territory, Cybernetics, Structural Grammar, Monoidal Categories, Semantic Web, Desiloification, Systemic Esperanto, Human-AI Collaboration

1.4 1. Introduction

1.4.1 1.1 The Problem of Disciplinary Silos and the “Desiloification” Hypothesis

Modern civilization confronts challenges — climate change, pandemic response, artificial intelligence governance, sustainable energy transitions — that are inherently systemic and transcend any single discipline. Yet our intellectual infrastructure remains fragmented. The nuclear engineer designing reactor safety systems, the photographer balancing the *Exposure Triangle*, the mythologist interpreting the Norse cosmological tree *Yggdrasil*, the nephrologist modeling the RAAS (Renin-Angiotensin-Aldosterone System), and the distributed systems researcher analyzing Nakamoto consensus in Bitcoin all grapple with structurally isomorphic challenges: maintaining equilibrium through balanced competing forces, managing trade-offs, designing for stability under perturbation, resisting entropy-driven degradation. Yet these practitioners have no shared language to recognize their kinship.

This fragmentation generates three compounding problems. First, **redundant reinvention**: *Negative feedback* (control theory), *Homeostasis* (biology), *Equilibration* (economics), and *Mean reversion* (finance) describe the same fundamental pattern, yet communities rarely cross-pollinate. Second, **integration barriers**: insights from hormonal cascade regulation in physiology go unrecognized as structural twins of neutron moderation in nuclear physics, or of proof-of-work difficulty adjustment in blockchain consensus. Third, **ontological proliferation**: without principled abstraction, knowledge systems explode with domain-specific concepts, becoming cognitively intractable.

The central hypothesis of TSCG — the *desiloification hypothesis* — is this: **most systems, whether natural or artificial, physical or abstract, share a set of generic, recurring, transdisciplinary principles that can be identified, formalized, and used as a shared vocabulary**. The project does not claim these principles are exhaustive or that they constitute a complete description of reality. It claims only that they are sufficiently common and structurally robust to serve as a practical construction kit for systemic modeling.

A corollary hypothesis — which TSCG's growing instance corpus progressively supports — is that the *same monoidal product formula* can describe structurally homologous phenomena across entirely unrelated domains. Dissipation ($F \times D$) governs energy loss in thermodynamics, signal attenuation in electronics, and hash-rate decay in a declining proof-of-work network alike. This cross-domain resonance is not metaphor: it is the empirical claim that TSCG invites the research community to scrutinize and challenge.

TSCG does not emerge from a void. Several prior frameworks have pursued structurally related ambitions. Stafford Beer's **Viable System Model** (VSM, 1972) demonstrated that the recursive structure of viable organizations could be described by a small set of cybernetic functions, independent of domain. Genrich Altshuller's **TRIZ** (1946–1985) distilled forty inventive principles from hundreds of thousands of patents, revealing that inventive solutions recur across engineering domains — a strikingly early empirical validation of the desiloification hypothesis. Alfred Korzybski's **map-territory distinction** (1933) provides TSCG's epistemological backbone, extended here into a formalized bidirectional feedback loop between representation and phenomenon. More broadly, TSCG inherits from **General System Theory** (von Bertalanffy, 1968), **Cybernetics** (Wiener, 1948), **Category Theory** (Eilenberg & Mac Lane, 1945), and **Semantic Web** ontology engineering (Berners-Lee et al., 2001).

Where TSCG differs from these precedents is in its *combinatorial* approach: rather than proposing a fixed set of domain-specific patterns or a single organizational archetype, TSCG constructs a vocabulary of atomic transdisciplinary concepts — **GenericConcepts** — that combine through monoidal products to generate domain-specific patterns on demand. The toolkit is deliberately open-ended and self-revising: a construction kit, not a finished cathedral.

This aspiration has a linguistic precedent. In 1887, Ludwig Lazarus Zamenhof proposed **Esperanto** as a constructed common language designed to transcend national barriers — not by replacing any natural language, but by offering a neutral shared medium. TSCG pursues an analogous ambition at the level of systemic knowledge: a **Systemic Esperanto**, a constructed common vocabulary for describing the structural behavior of systems across disciplines. The goal is not *desiloification by erasure* but *desiloification by translation* — providing a shared layer of abstraction through which practitioners from biology, engineering, economics, mythology, and blockchain research can recognize structural kinship without abandoning their own domain vocabulary.

TSCG makes no claim to be a new Theory of Everything. Its 80 GenericConcepts do not exhaust the structure of reality; its monoidal formulas are not laws of nature. The vocabulary offered here involves deliberate, partially arbitrary choices — submitted, not imposed.

1.4.2 1.2 Origin: Twenty-Five Years of Intuition, Then AI-Assisted Concretization

This toolkit did not emerge from an academic program. It grew from more than **twenty-five years** of informal, persistent reflection — what the author describes as “creative meditation” — on whether generic principles truly recur across most systems. What spanned those years was not the ASFID/REVOI dimensions themselves, but the underlying intuition: the hypothesis that transdisciplinary structural invariants exist and can be named.

The concretization of this intuition is recent (approximately 5–6 months of active development as of June 2026). It began with approximately twenty exploratory conversations with DeepSeek, during which the ASFID dimensions first emerged in recognizable form. This was followed by sustained collaboration with Claude AI (Anthropic), which became the primary and lasting partner for all subsequent work: deepening the framework, formalizing the M3 foundation, developing the ontology layers, and growing the instance corpus.

The decisive infrastructure step came with Claude's **Projects** feature, which maintains a persistent corpus of reference documents across sessions — including the **Smart Prompt**, a structured document summarizing all key architectural decisions, naming conventions, and project state. This enabled systematic formalization, publication on GitHub, and iterative development of the instance corpus.

1.4.2.1 Delegated tasks (Claude AI as executor)

Algebraic formalization. The monoidal product formulas characterizing each GenericConcept, and the Structural Grammar foundation (Lambek calculus, free commutative monoidal categories) were formulated by Claude AI on the basis of the author's conceptual specifications. These formulations are acknowledged as *AI-assisted approximations* pending rigorous mathematical review.

Documentary research per instance. For each instance under development, Claude AI conducted targeted literature research — sourcing domain-specific descriptions, identifying key structural components, and synthesizing cross-disciplinary analogies.

JSON-LD ontology encoding. Translation of conceptual models into well-formed JSON-LD files — respecting W3C standards, TSCG namespace conventions, and ontological consistency constraints — was delegated to Claude AI. The author reviewed and validated each generated file.

1.4.2.2 Collaborative tasks (Claude AI as active partner)

GenericConcept identification and modeling. The identification of candidate GenericConcepts — particularly during instance analysis, when a new system reveals structural patterns not yet captured in M2 — is a genuinely collaborative process. The author proposes candidates based on systemic intuition; Claude AI evaluates monoidal decomposition, checks consistency with existing M2 concepts, and proposes formulations.

Instance simulation generation. Each standalone HTML simulation is generated collaboratively. Claude AI takes initiative on illustrative mechanisms; the author steers ergonomics and pedagogical credibility through iterative feedback.

Workflow engineering via Skills pipelines. Two structured AI *Skills* — reusable pipeline specifications encoding the author's accumulated methodology — have been formalized: an **Instance Analysis Pipeline** (proposal → analysis → modeling → simulation) and a **Research Article Pipeline** (audit → planning → drafting → revision → finalization).

1.4.2.3 Continuity infrastructure

The collaboration is sustained across sessions by two complementary mechanisms. The **Smart Prompt** re-injects architectural state at the start of each new session. The **Skills** pipeline system encodes procedural knowledge — not what the toolkit is, but how to work with it — as reusable, versioned AI instructions.

1.4.3 1.3 The Bicephalous Architecture: Stereoscopic Knowledge

To understand TSCG's architecture, consider a creature from Mesopotamian mythology: **Anzû**, a divine entity depicted as a lion-headed eagle — simultaneously earthly and celestial, interpretive and calculative. In TSCG, Anzû serves as the foundational metaphor for the M3 layer: a single toolkit with two heads, each possessing one eye.

The critical detail is the *one eye per head*. In optics, a single eye produces **monocular vision**: a flat, two-dimensional projection with no depth perception. Two eyes, positioned at a slight angle to each other, produce **binocular vision**: the brain triangulates the two slightly different images into a three-dimensional depth map. The same principle governs epistemic architecture.

The Eagle Head scrutinizes the world as it *is* — measuring, instrumenting, quantifying phenomena. Its eye, the **Eagle Eye**, operates through the five ASFID dimensions: Attractor, Structure, Flow, Information, Dynamics. It asks of any system: what draws it toward equilibrium? what organizes it? what flows through it? what information governs it? what dynamics evolve it? The Eagle Eye produces a Territory score vector — a quantified description of the system's observable behavior.

The Sphinx Head contemplates the world as it is *represented* — modeling, theorizing, interpreting. Its eye, the **Sphinx Eye**, operates through the five REVOI dimensions: Representability, Evolvability, Verifiability, Observability, Interoperability. It asks of any model: can this system be adequately represented? can the model evolve as the system does? can its predictions be verified? can its behavior be observed? can it communicate with other models? The Sphinx Eye produces a Map score vector — a quantified assessment of the model's epistemic quality.

Neither head can function alone. The Eagle Head without the Sphinx Head produces raw measurements with no interpretive framework — data without meaning. The Sphinx Head without the Eagle Head produces elegant theories with no empirical grounding — meaning without data. Each head, alone, has monocular vision: a flat, depthless perspective. Together, they produce **stereoscopic epistemic depth**: the ability to perceive not only the system, but the *distance* between our model of it and the reality it represents.

This distance is formalized as the **epistemic gap** δ_1 :

$$\delta_1 = |\text{ASFID_mean} - \text{REVOI_mean}| / \sqrt{2}$$

A small δ_1 indicates that the model adequately tracks the system (SpectralClass: Coherent); a large δ_1 signals either model inadequacy or an under-observed Territory (SpectralClass: Enigmatic).

But two-eyed stereopsis, however powerful, still has limits: it measures *distance* between Map and Territory, but not the *quality of their fusion*. This is where the third grammar enters.

The Stereopsis Grammar (Gs / TKSL) is TSCG's most recent architectural addition. Where the Eagle Eye (Gt) and the Sphinx Eye (Gm) each operate within their own monoidal grammar, Gs formalizes the act of *triangulation itself* — the stereopsis focal point where Territory and Map converge into actionable knowledge. Its four nominal primitives are: **T** (Temporality — *when?*), **K** (Knowledge — *what?*), **Ss** (Symbol — *what sign?*), and **L** (Localizability — *where does it converge?*). Two polar modifiers — ****_**** (PositivePole, onset/amplifying) and ****_\$\$\$** (NegativePole, terminus/attenuating) — are not primitives but modifiers governing polarity within Gs expressions.

The quality of stereopsis fusion is measured by the **Epistemic Focal Score (EFS / δ_2)**:

$$\delta_2 = \text{stereopsisDepth} \times (1 - |\text{focalBias}|)$$

where `stereopsisDepth` reflects the proportion of active TKSL primitives mobilized by the instance, and `focalBias` (`(REVOI_mean - ASFID_mean)`) indicates whether the model leans toward Map abstraction (Hyperopic) or Territory detail (Myopic). A balanced, deep stereopsis yields a high δ_2 (FocalClass: Emmetropic); a collapsed fusion yields a low δ_2 (FocalClass: Astigmatic).

Together, the three grammars — Gt (Eagle Eye / Territory), Gm (Sphinx Eye / Map), Gs (Stereopsis fusion) — constitute the **Base16**: 16 primitives in total (13 typed + 3 neutral operators), forming the complete generative vocabulary of the TSCG M3 foundation.

Figure 1 illustrates this architecture as a hexagonal monoidal grid. Red hexagons represent Gt/ASFID Territory primitives; green hexagons represent Gm/REVOI Map primitives; blue hexagons represent Gs/TKSL Stereopsis primitives. The connecting lines — traversing all three grammars — illustrate how a *GenericConcept* simultaneously mobilizes primitives from multiple grammars, producing a cross-grammar monoidal expression.

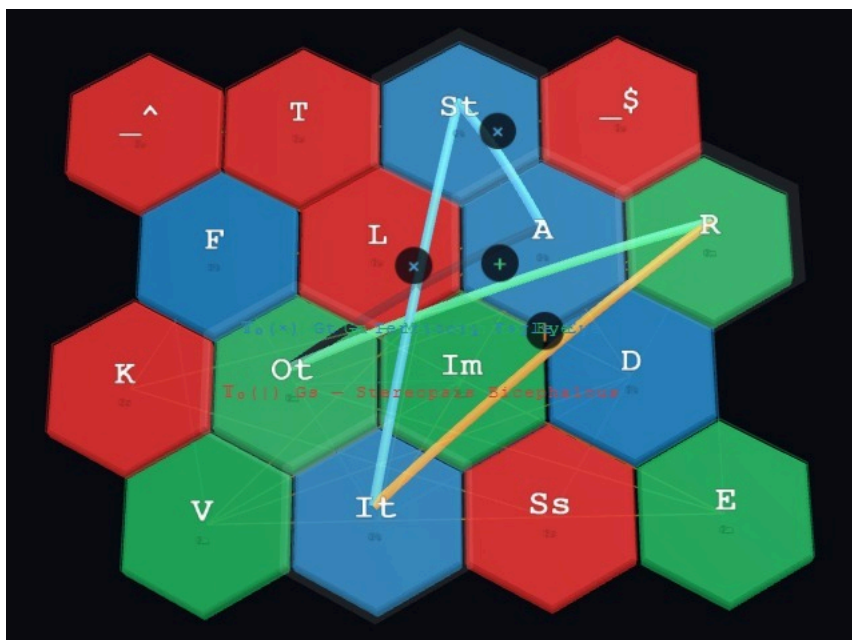


Figure 1 — TSCG Base16 Hexagonal Monoidal Grid

Figure 1 — The TSCG Base16 hexagonal monoidal grid. Red hexagons: Gt/ASFID Territory primitives (A, S, F, It, D). Green hexagons: Gm/REVOI Map primitives (R, E, V, O, Im). Blue hexagons: Gs/TKSL Stereopsis primitives (T, ^, \$, K, Ss, L). The connecting lines illustrate how a GenericConcept simultaneously mobilizes primitives from multiple grammars — here, the $\times$ (Territory) and $+$ (Map) operators — producing a cross-grammar monoidal expression. The two operator nodes (dark circles labeled $\times$ and $+$) mark the monoidal fusion points.

1.5 2. The TSCG Architecture

TSCG is organized as a **four-layer hierarchical ontology**, implemented in JSON-LD following W3C semantic web standards. The layers are ordered from the most abstract (M3) to the most concrete (M0), connected by three functors that progressively instantiate and specialize the toolkit’s vocabulary:

M3 GenesisGrammar	— Mathematical foundation (3 monoidal grammars, Base16)
F_grammaticalization ↓	
M2 GenericConcepts	— 80 atomic transdisciplinary patterns (9 families)
F_instantiation ↓	
M1 Domain Extensions	— Knowledge-field vocabularies (Biology, Chemistry, Music...)
F_concretize ↓	
M0 Instances	— Validated models of concrete systems (33 instances)

Each layer is a strict refinement of the one above: M2 concepts are monoidal products of M3 primitives; M1 concepts are instantiations of M2 concepts in specific domains; M0 instances are concrete realizations of M1 concepts in specific systems. No layer introduces vocabulary that belongs to a higher layer — a discipline enforced by the **M2 Purity Principle** (§2.8).

1.5.1 2.1 M3 — The GenesisGrammar: Three Monoidal Grammars

The foundation of TSCG is the **GenesisGrammar** (`M3_GenesisGrammar.jsonld`): a set of three free commutative monoidal grammars whose primitives jointly constitute the complete generative vocabulary of the toolkit. This foundation — established in v16.x through the **Structural Grammar** migration — is grounded in Lambek calculus and category theory (Mac Lane, 1971).

Each grammar has its own operator, its own neutral element, and its own set of typed primitives. The operator alone identifies the grammar: no prefix is needed. Operator priority follows arithmetic convention: $\times$ binds tighter than $+$, which binds tighter than $|$. Thus `A  $\times$  S + R | K` reads as `((A  $\times$  S) + R) | K`.

Grammar	Operator	Neutral element	Acronym	Primitives	Count
Gt (Territory)	$\times$	EmptyTerritory	ASFID	A, S, F, It, D	5
Gm (Map)	$+$	EmptyMap	REVOI	R, E, V, O, Im	5
Gs (Stereopsis)	$\backslash $	EmptyStereopsis	TKSL	T, ^, \$, K, Ss, L	6

Total: Base16 — 16 primitives (13 typed + 3 neutral operators).

A GenericConcept such as Homeostasis is written `A  $\times$  St  $\times$  F | L` — a monoidal expression whose meaning emerges from the simultaneous, non-separable co-presence of its primitives. This requires no metric, no inner product, and no notion of orthogonality — only the algebraic structure of a free commutative monoidal category.

1.5.2 2.2 Gt — The Eagle Eye (Territory / ASFID)

The Territory grammar **Gt** measures systems as they empirically exist. Its five primitives decompose any system along independent dimensions:

Primitive	Name	Operational question
A	Attractor	What draws the system toward equilibrium or stable states?
S	Structure	What stable organizational pattern governs its components?
F	Flow	What transfers energy, matter, or information through it?
It	Information	What signals, codes, or cybernetic loops govern its behavior?
D	Dynamics	How does it evolve temporally through state transitions?

Each primitive is scored on [0, 1] during instance analysis, producing the **ASFID score vector** that quantifies the Territory footprint of the system.

⚠ **It** (Information, Territory) is systematically distinguished from **Im** (Interoperability, Map) in all formulas and score properties (`m0:scoreIt` vs `m0:scoreIm`).

1.5.3 2.3 Gm — The Sphinx Eye (Map / REVOI)

The Map grammar **Gm** evaluates models as epistemic constructs. Its five primitives assess the quality of any representation of a system:

Primitive	Name	Operational question
R	Representability	Can this system be adequately encoded in a formal model?
E	Evolvability	Can the model update as the system changes?
V	Verifiability	Can the model's predictions be empirically tested?
O	Observability	Can the system's relevant states be perceived and measured?
Im	Interoperability	Can this model integrate with and translate to other models?

⚠ **R = Representability** (semantic encodability/decodability) — never “Reproducibility”.

REVOI scores are inherently **observer-relative**: the same system may receive different REVOI scores from a biologist and an engineer. This relativity is a feature, not a defect — the Map is always the map of *someone*.

1.5.4 2.4 Gs — The Stereopsis Grammar (TKSL)

The Stereopsis grammar **Gs** is TSCG's most architecturally distinctive layer. Where Gt measures the Territory and Gm constructs the Map, **Gs formalizes the act of triangulation itself** — the focal convergence where Territory observation and Map construction produce actionable, situated knowledge.

Gs operates with the `|` operator (stereopsis fusion):

Element	Type	Name	Question	Theoretical basis
T	Primitive	Temporality	<i>When?</i>	Temporal interface Gt↔Gm
K	Primitive	Knowledge	<i>What?</i>	Maturana/Varela cognition
Ss	Primitive	Symbol	<i>What sign?</i>	Peirce, Saussure
L	Primitive	Localizability	<i>Where does it converge?</i>	Wiener, Ashby
_^	Modifier	PositivePole	<i>(onset/amplifying)</i>	Polarity modifier
\$	Modifier	NegativePole	<i>(terminus/attenuating)</i>	Polarity modifier

Two special derived elements:

- **Pole Annihilation Axiom:** `_^ | _$` = `StereopsisEmptySet` — named constant `#Null`. When positive and negative poles cancel, stereopsis fusion collapses.
- **TriadicBalance:** `_0 = _^ | _$` — the central equilibrium state of Gs, a `DerivedGsElement` preserving Base16 integrity.

Notation for hybrid formulas: when a formula spans multiple grammars, the operator identifies the grammar unambiguously:

```
Pure Territory:  A × S × F
Pure Map:       R + E + V
Hybrid:         A × St × It × D | K
                ^^^^^^^^^^^^^^^^^ ^
                Gt primitives    Gs primitive
```

In hybrid formulas, Territory and Map primitives carry their monoid index (St, It) to avoid collision with Gs primitives sharing similar names.

Figure 1 illustrates the Base16 as a hexagonal monoidal grid: red hexagons = Gt/ASFID; green hexagons = Gm/REVOI; blue hexagons = Gs/TKSL. The connecting lines show how a *GenericConcept* simultaneously mobilizes primitives from multiple grammars, producing a cross-grammar monoidal expression.

1.5.5 2.5 M2 — GenericConcepts: The Transdisciplinary Vocabulary

M2 hosts **80 atomic GenericConcepts** organized into 9 families. Each *GenericConcept* carries a monoidal formula encoding its structural signature as a product of M3 primitives. Full catalog in Appendix C; architecture detailed in §3.

1.5.6 2.6 M1 — Domain Extensions

M1 instantiates M2 concepts within specific knowledge fields. Thirteen active extensions cover Biology, Chemistry, Physics, Electronics, Economics, Photography, Music, Mythology, Optics, SystemicModeling, Education, Geology, and BusinessModeling. Each M1 extension defines **KnowledgeFieldConceptCombos** — domain-specific combinations of *GenericConcepts* validated within a single knowledge field.

1.5.7 2.7 M0 — Instances

M0 is the empirical layer: **33 validated instances** implementing M1/M2 concepts against concrete systems. The instance types and their counts are detailed in §4.

1.5.8 2.8 The Purity Principle

A discipline enforced at every layer: **no concept belongs to a layer lower than its degree of generality warrants**. A concept earns M2 membership only by demonstrating structural relevance across at least six unrelated domains (the six-domain threshold, §6.2). Domain-specific concepts belong in M1; concrete system models belong in M0.

The anti-pattern — **Ontological Overfitting** — occurs when a concept is added to M2 because it appears in one well-studied system, creating a spurious 1:1 map-territory correspondence. TSCG actively resists this pressure, documented in `OntologicalOverfitting.md`.

1.6 3. The GenericConcepts Layer (M2)

1.6.1 3.1 Structural Grammar as a Concept Construction Language

M2 GenericConcepts are not defined by verbal description alone — each carries a **monoidal formula** encoding its structural signature as a product of M3 primitives. The formula is written using the operators of the relevant grammar(s): $\times$ for Territory (Gt), $+$ for Map (Gm), $|$ for Stereopsis (Gs), with priority $\times > + > |$.

The formula captures the idea that a GenericConcept requires the *simultaneous, non-separable co-presence* of its primitives. Homeostasis ($A \times St \times F | L$) cannot be reduced to Attractor alone, nor to Structure alone — it emerges only from their joint activation. This is not a metaphor: it is a structural claim, expressed in the language of free commutative monoidal categories (Lambek calculus), that the concept is irreducible to any proper subset of its primitives.

Four morphism types relate GenericConcepts to each other:

- Inclusion** $\hookrightarrow$: Homeostasis $\hookrightarrow$ Regulation (specialization)
- Composition** $\circ$: Learning = Memory $\circ$ Adaptation
- Duality** $\wedge^{op}$: Convergence^{op} = Divergence; Entropy^{op} = Negentropy
- Emergence** $\Rightarrow$: $M_A \times M_B \Rightarrow M_C$ (GenericConceptCombo)

1.6.2 3.2 The Nine Families

GenericConcepts are organized into nine families reflecting their dominant structural role. The complete listing with monoidal formulas is in Appendix C; here we present each family with representative examples.

Structural (23 concepts): Patterns governing the stable organization of components. *Examples:* Network ($S \times I \times F | O + R + Im + V$), Hierarchy ($S \times A$), Coherence ($A \times St \times It | R + O | _{}^{\wedge} / _{}^{\$}$).

Dynamic (17 concepts): Patterns governing temporal transformation and process. *Examples:* Process ($D \times F$), Bifurcation ($D \times F | L$), Modelisation ($D \times F \times It | R + V + E$), TriadicBalance ($A \times F \times D | _{}^{\theta}$).

Ontological (10 concepts): Patterns governing systemic environment and framing. *Examples:* Context ($O + R + Im + E$), Observer ($I \times A$), State (I).

Regulatory (10 concepts): Patterns governing stability and control. *Examples:* Homeostasis ($A \times St \times F | L$), Regulation ($A \times S \times F | R + V + E$), Threshold ($A \times I$).

Informational (8 concepts): Patterns governing encoding, signaling, and representation. *Examples:* Signal ($I \times F$), Code ($It \times Ss$), Language ($Ss \times F | K$).

Adaptive (4 concepts): Patterns governing learning and resilience. *Examples:* Adaptation ($I \times F \times D$), Memory ($\int (D-F) dt$), Resilience ($A \times S$).

Energetic (4 concepts): Patterns governing energy transformation and dissipation. *Examples:* Dissipation ($F \times D$), Entropy ($F \times It \times D | _{}^{\wedge} / _{}^{\$}$), Transducer ($F \times S \times I$).

Relational (5 concepts): Patterns governing connections and interactions. *Examples:* Mediator ($F \times It | K$), Agent ($St \times It \times D | K$), Role ($Ss | K$).

Teleonomic (1 concept): Self-Organization ($A \times I \times D$).

1.6.3 3.3 Bicephalous GenericConcepts

A subset of GenericConcepts span *both* Gt and Gm grammars in a single formula. These **bicephalous GenericConcepts** capture phenomena that are simultaneously territorial (observable) and cartographic (representational) by nature.

The most architecturally significant example is **Coherence** ($A \times St \times It | R + O | _{}^{\wedge} / _{}^{\$}$):

- $A \rightarrow$ alignment toward a common attractor (Territory / Eagle Eye)
- $St \rightarrow$ globally continuous structure (Territory / Eagle Eye)
- $It \rightarrow$ shared / compatible information (Territory / Eagle Eye)
- $R \rightarrow$ Representability — coherence as modeled (Map / Sphinx Eye)
- $O \rightarrow$ Observability — coherence as perceived (Map / Sphinx Eye)

Note the St/It indexation: in this hybrid formula, Territory primitives S and I carry their monoid index (St, It) to avoid ambiguity with Map or Gs primitives.

Coherence admits four regimes (quaternary polarity):

	R + O strong	R + O weak
A $\times$ St $\times$ It strong	Verified truth	Misunderstood reality
A $\times$ St $\times$ It weak	Illusion / ideology	Total chaos

Two v16.2.0 additions are notably bicephalous or cross-grammar:

- Modelisation** ($D \times F \times It | R + V + E$): the act of constructing a formal representation — simultaneously a structural operation (Gt) and an epistemic assessment (Gm). Transdisciplinary across scientific modeling, cartography, software architecture, legal codification, musical notation, and linguistics.
- TriadicBalance** ($A \times F \times D | _{}^{\theta}$): equilibrium achieved through the triangulation of three perspectives. The $_{}^{\theta}$ (EquilibriumPole) is the qualitative optimum — not an arithmetic midpoint — toward which the system is attracted.

1.6.4 3.4 GenericConceptCombos

Beyond atomic GenericConcepts, M2 defines the **GenericConceptCombo** class for synergistic combinations producing emergent patterns not reducible to their components. Instances of GenericConceptCombo are defined in M1_CoreConcepts.jsonld. TSCG v16.2.0 defines four validated Combos:

Combo	Formula	Components	Polarity
Inertia	$S \times F \times I \times D$	$\Rightarrow$ (Memory, Entropy)	Neutral
Potentialization	$A \times D \times F \mid _{}^{\wedge} / _{}^{\$}$	$\Rightarrow$ (Activation, Process)	Dual
Absorbing State	$S \times A \times F \times I \times D$	$\Rightarrow$ (Stase, Entropy)	Neutral
Topological Defect	$St \times A \times It \mid R + O$	$\Rightarrow$ (Incoherence, Invariant)	Neutral

Potentialization resolves a long-standing tension: concepts like Resource, Storage, Hub, and Stateful — previously problematic because they implied F=0 — are now understood as systems in suspended Potentialization (F_potential mode).

Absorbing State is a Stase sealed irreversibly by Entropy production — a terminal state from which Potentialization is structurally impossible, in contrast to ordinary Stase ($S \times A$) which remains reversible by construction (D absent).

1.6.5 3.5 The Causal Chain for Irreversibility

The analysis of the NakamotoConsensus instance produced the most significant single M2 expansion in the toolkit’s history: five new atomic GenericConcepts and a foundational M3 amendment (Flow axiom relaxation: $F \geq 0$).

The five additions:

Concept	Formula	Family	Key insight
Processor	$S \times I \times F \times D$	Dynamic	Generic transformer
Transducer	$F \times S \times I$	Energetic	Converts flux type; subClassOf Processor
Entropy	$F \times It \times D \mid _{}^{\wedge} / _{}^{\$}$	Energetic	Irreversibility measure
Stase	$S \times A$	Structural	F=0 reversible ground state
Coherence	$A \times St \times It \mid R + O \mid _{}^{\wedge} / _{}^{\$}$	Structural	Bicephalous

Together these constitute a **causal chain for irreversibility**:

F_active (available exergy)

↓ Dissipation ($F \times D$) — equations time-reversible (Feynman)

F_degraded → Entropy ($F \times It \times D \mid _{}^{\wedge} / _{}^{\$}$) ← irreversibility emerges here

↓ Memory ($\int dt$)

Inertia $\Rightarrow$ (Memory, Entropy)

↓ maximal → seals Stase

Absorbing State $\Rightarrow$ (Stase, Entropy)

The Energetic hierarchy is now:

Processor ($S \times I \times F \times D$)

└ Transducer ($F \times S \times I$) subClassOf Processor

└ Dissipation ($F \times D$) subClassOf Transducer

└ produces → Entropy ($F \times It \times D \mid _{}^{\wedge} / _{}^{\$}$)

1.6.6 3.6 M2 Purity and the Six-Domain Threshold

A candidate GenericConcept earns M2 membership only by demonstrating structural relevance across **at least six unrelated knowledge fields**. This threshold guards against Ontological Overfitting — the anti-pattern of adding a concept to M2 because it appears prominently in one well-studied system.

The NakamotoConsensus instance provided the sixth-domain validation for Dissipation and Entropy: confirming their M2 status by demonstrating structural relevance in distributed consensus alongside thermodynamics, biology, electronics, photography, and mechanical engineering.

1.6.7 3.7 Mathematical Validation Status

The monoidal formulas in this section were developed collaboratively with Claude AI and carry the ontology’s own assessment: *preliminary, AI-assisted, pending expert review*. The planned path toward formal validation is SHACL encoding — expressing monoidal product constraints as machine-verifiable SHACL shapes — identified as a priority for a future release.

The Structural Grammar foundation (Lambek calculus, free commutative monoidal categories) provides a more rigorous algebraic basis than the earlier Hilbert-space approximation. Nevertheless, the claim that TSCG’s $\times$ operator satisfies all axioms of a free commutative monoidal category in the strict mathematical sense remains to be verified by a domain expert.

1.7 4. The Instance Corpus

1.7.1 4.1 Instances as the Empirical Backbone of TSCG

The primary empirical instrument of TSCG is the **instance** — a validated model of a concrete system encoded as a JSON-LD ontology file (`M0_*.jsonld`), structured according to the TSCG four-layer architecture and constrained by a SHACL grammar (`M0_Instances_Schema.shacl.ttl`).

An instance is not a simulation, nor a description. It is a **formal claim**: that a given system’s structure can be adequately characterized by a combination of GenericConcepts drawn from M2, instantiated through M1 domain extensions, and scored against the ASFID/REVOI dimensions. Each instance is falsifiable: a well-formed counter-argument (demonstrating that a key GenericConcept is absent, or that a score is systematically miscalibrated) constitutes a valid challenge to the instance’s modeling claims.

Historically, the first instance type to emerge was the **Poclet** — a minimal, self-contained model of a single system, designed for pedagogical clarity and structural transparency. The corpus subsequently extended to three additional instance types as the toolkit matured:

Poclet	– Minimal model of a single concrete system
SystemicFramework	– Re-encoding of an existing formal framework in TSCG
SymbolicSystemGrammar	– Encoding of a symbolic/generative system
TscgTool	– Software instrument operating reflexively on the toolkit

All four types share the same JSON-LD structure, SHACL constraints, and scoring conventions. They differ in their **ontological role** (`m3:ontologyType`) and in the nature of the system being modeled.

As of v16.2.0, the corpus contains **33 validated instances** across all four types.

1.7.2 4.2 Poclets (25 validated)

A Poclet (from *pocket* + *let*) is the foundational instance type: a minimal, self-contained TSCG model of a single system. The name reflects the design philosophy — small enough to fit in a pocket, complete enough to stand alone.

Poclets span twelve knowledge domains, demonstrating the transdisciplinary reach of the GenericConcept vocabulary:

Domain	Poclet	Key GenericConcepts
Biology	AdaptativeImmuneResponse	Homeostasis, Memory, Threshold
Biology	BloodPressureControl	Regulation, Homeostasis, Feedback
Biology	ButterflyMetamorphosis	Process, Transformation, Threshold
Biology	CellSignalingModes	Signal, Code, Transducer
Biology	ComplexChemicalSynapse	Signal, Threshold, Transducer
Biology	Kidneys	Homeostasis, Filter, Gradient
Biology	Raas	Regulation, Cascade, Homeostasis
Biology	TrophicPyramid	Hierarchy, Flow, Dissipation
Chemistry	PhaseTransition	Bifurcation, Threshold, Stase
Distributed Systems	NakamotoConsensus	Coherence, Entropy, Dissipation
Economics	KindlebergerMinsky	Bifurcation, Cycle, Attractor
Electronics	Transistor	Threshold, Transducer, Switch
Electronics	Vco	Regulation, Oscillation, Signal
Energy	FourStrokeEngine	Cycle, Dissipation, Transducer
Energy	FireTriangle	Threshold, Flow, Activation
Gaming	MtgColorWheel	Symmetry, Constraint, Network
Geology	PlateTectonics	Flow, Gradient, Bifurcation
Music	Counterpoint	Constraint, Regulation, Pattern
Norse Mythology	Yggdrasil	Network, Hierarchy, Axis Mundi
Nuclear Engineering	NuclearReactorsTypology	Regulation, Threshold, Transducer
Optics / Color	ColorSynthesis (<i>fédérateur</i>)	Symmetry, Constraint, Convergence
Pedagogy	Tpack	Network, Modularity, Intersection
Photography	ExposureTriangle	Regulation, Constraint, Gradient
Physics	Ptoe	Hierarchy, Symmetry, Invariant
Signal Processing	TvTestPattern	Code, Signal, Pattern

Note: *ColorSynthesis* is a federated Poclet aggregating sub-ontologies (e.g., *M0_RGB_Additive.jsonld*, *M0_CMY_Subtractive.jsonld*).

1.7.3 4.3 SystemicFrameworks (3 validated)

A **SystemicFramework** re-encodes an existing formal framework — already possessing its own internal structure and vocabulary — through the TSCG lens.

VSM — **Viable System Model** (Beer, 1972): The recursive cybernetic model of viable organizations is re-encoded as a five-level Regulation/Homeostasis hierarchy, with System 3★ (algedonic channel) modeled as a Threshold-triggered Signal.

TRIZ — **Theory of Inventive Problem Solving** (Altshuller, 1946–1985): The 40 inventive principles and contradiction matrix are re-encoded as a combinatorial Constraint × Transformation space, with the Ideal Final Result modeled as an Attractor.

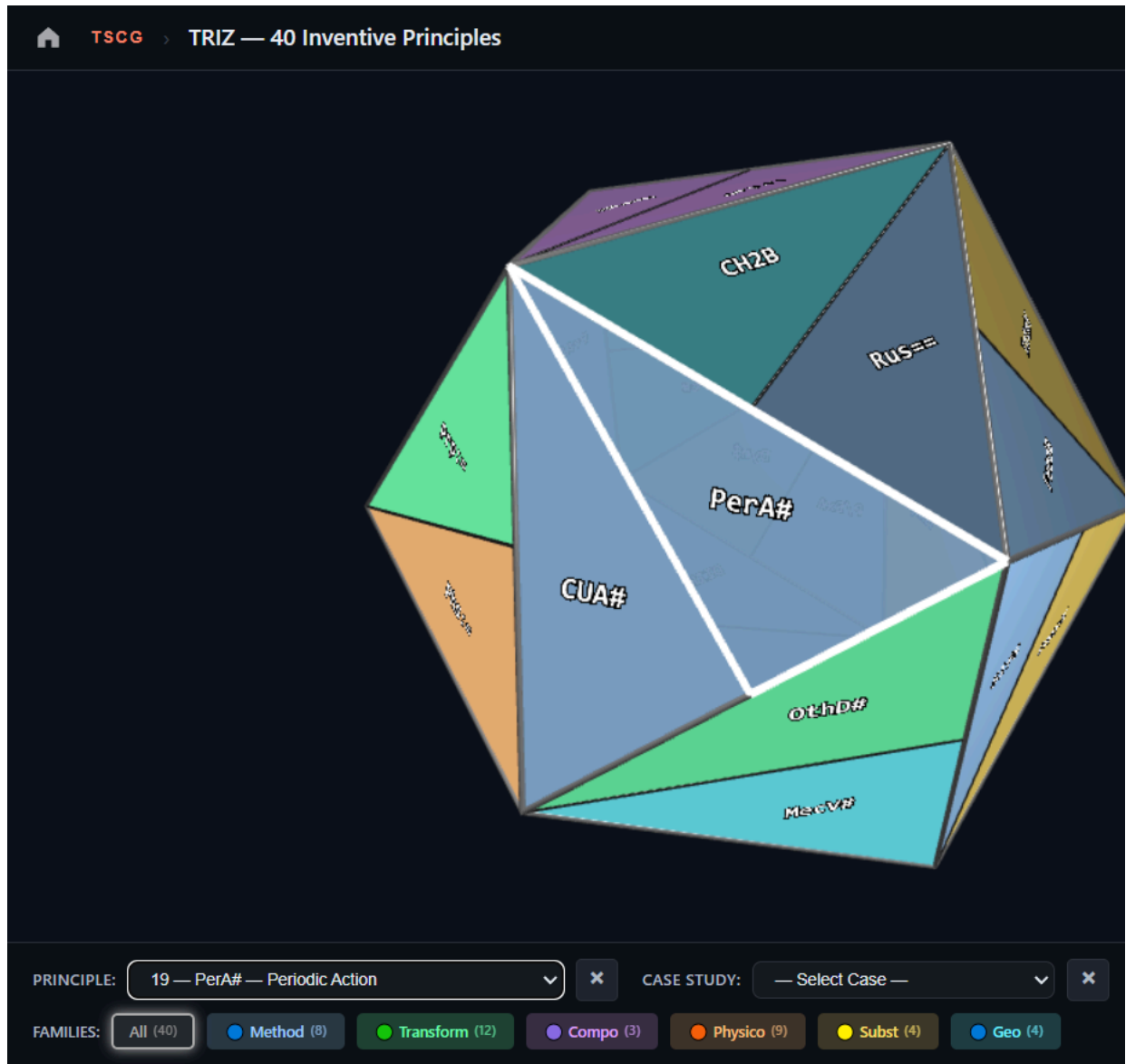


Figure 2 — TRIZ SystemicFramework BabylonJS simulation

Figure 2 — The TRIZ SystemicFramework BabylonJS 3D simulation. The 40 inventive principles are distributed across the faces of a polyhedron, color-coded by family (Method, Transform, Composition, Physics, Substance, Geometry). The interactive interface allows navigation by principle, family, and case study, exposing the structural topology of the TRIZ contradiction space as a TSCG Constraint × Transformation manifold.

BMC — **Business Model Canvas** (Osterwalder, 2004): The nine BMC blocks are re-encoded across ASFID dimensions. This SystemicFramework has the most advanced simulation: 12 real-world company cases (Netflix, Nokia, Nintendo, Apple, Amazon, Google, Microsoft, IBM, Kodak, Xerox, Michelin, Airbnb), 5 lifecycle phases, and 34 documented transitions, rendered as an interactive p5.js Canvas2D simulation.

Note: BMC has two complementary ontology files. `M0_Bmc.jsonLd` is the TSCG instance model; `M0_BmcSimulation.jsonLd` encodes the use-case data (12 companies, 34 transitions) that feeds the interactive simulation.

1.7.4 4.4 SymbolicSystemGrammars (2 validated)

A **SymbolicSystemGrammar** encodes a symbolic or generative system whose primary reality is semiotic rather than physical.

I-Ching: The 64 hexagrams are re-encoded as a binary combinatorial system ($2^6 = 64$ states) built from a Symmetry × Code × Cycle substrate. The I-Ching constitutes a 3,000-year-old empirical proof of the desiloification hypothesis: a culture-independent structural vocabulary for systemic states predating modern systems theory by millennia.

TriskeleToolchain: A custom Rust toolchain (tsk-cc, tsk-link, triskele-vm, tsk-asm) compiling C via LLVM IR to bytecode for a custom RISC virtual machine. Its ISA (Instruction Set Architecture) comprises **256 opcodes organized into exactly 16 categories** — one per M3 primitive, mapped bijectively to the Base16. This constitutes an independent structural validation of the Base16 from a completely unrelated domain: virtual machine design and compiler engineering (see §6.4).

1.7.5 4.5 TscgTools (3 validated)

TscgTools are software instruments that operate **reflexively** on the toolkit itself — instances whose domain is TSCG’s own infrastructure.

TscgOntologyExplorer: Desktop application (ElectronJS) for navigating the TSCG ontology hierarchy.

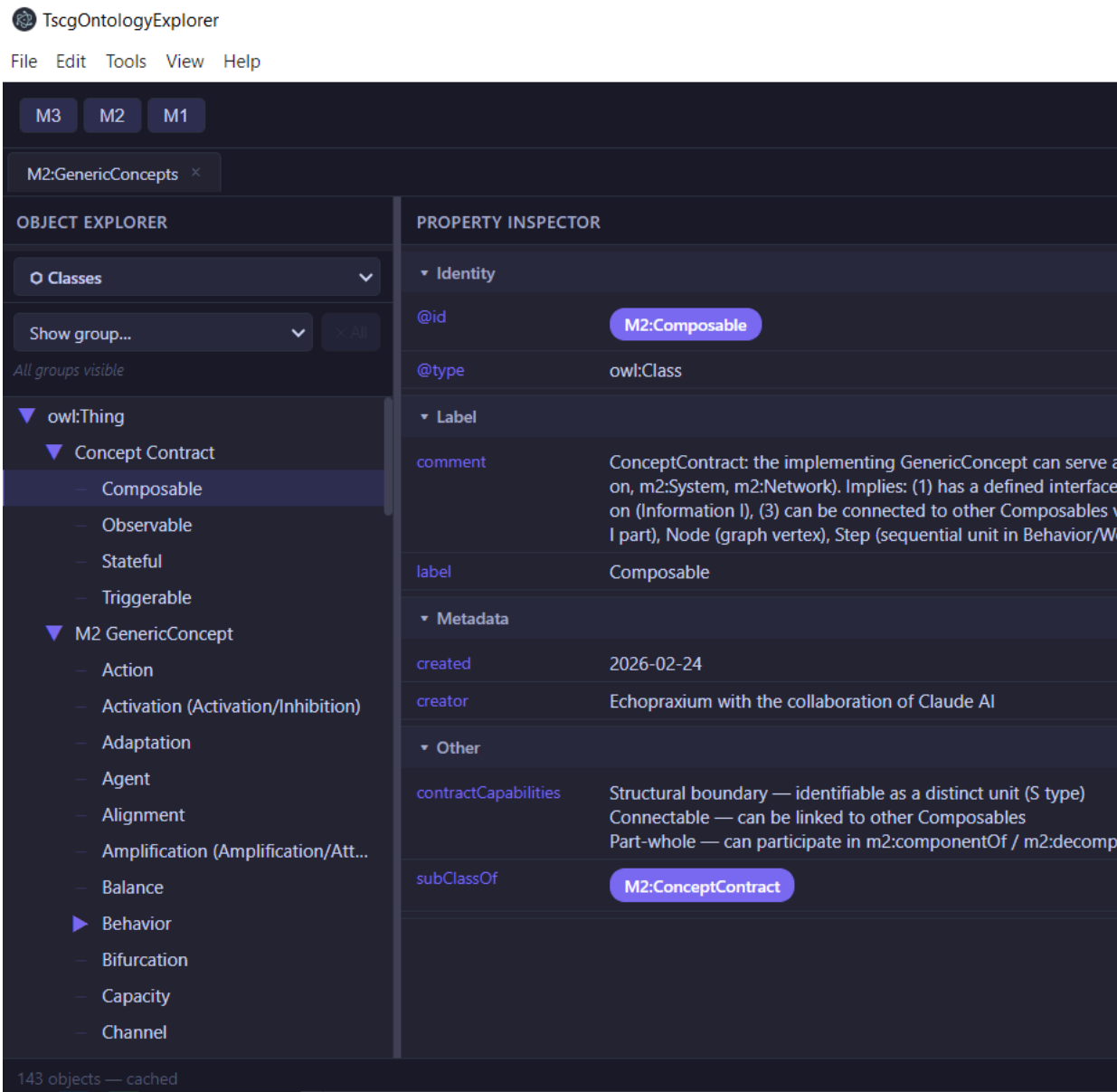


Figure 3 — TscgOntologyExplorer desktop application

Figure 3 — The TscgOntologyExplorer desktop application (ElectronJS, v1.0). The three-panel interface exposes M3/M2/M1 layer navigation (left: Object Explorer with class hierarchy), property inspection (center: Property Inspector showing @id, @type, label, creator, subClassOf), and SPARQL query execution (right: predefined queries including “All GenericConcepts”, “Structural Grammar Formulas”, “ASFID Scores of an instance”). Note the `creator: “Echopraxium with the collaboration of Claude AI”` field — the authorship convention is machine-readable and enforced by SHACL.

TscgOntologyAPIServer: Local REST API server exposing the TSCG ontology for programmatic access.

TscgOntologyAPIServer

1.1.0 OAS 3.1

/openapi.json

Standalone REST server exposing the TSCG ontology corpus via SPARQL 1.1 and corpus management endpoints. Uses pyoxigraph as a named-graph triple store (/docs).

Server

GET	/health	Health
GET	/info	Info

Corpus

POST	/corpus/load	Corpus Load
POST	/corpus/load-pattern	Corpus Load Pattern
GET	/corpus/list	Corpus List
GET	/corpus/layers	Corpus Layers
POST	/corpus/reload	Corpus Reload
POST	/corpus/reload-all	Corpus Reload All
DELETE	/corpus/graph	Corpus Delete Graph
DELETE	/corpus/clear	Corpus Clear
GET	/corpus/export	Corpus Export

SPARQL

Figure 4 — TscgOntologyAPIServer Swagger UI

Figure 4 — The TscgOntologyAPIServer Swagger UI (v1.1.0, OAS 3.1). The standalone REST server exposes the TSCG ontology corpus via SPARQL 1.1 and corpus management endpoints, using pyoxigraph as a named-graph triple store with optional disk persistence. Endpoint groups: Server (health, info), Corpus (load, list, layers, reload, export), SPARQL. This TscgTool demonstrates the reflexive character of the M0 layer: TSCG models itself as a queryable knowledge infrastructure.

TscgPocletMiner: AI-assisted instance generation tool supporting multiple LLM backends (Claude AI, Gemini, DeepSeek, Ollama).

1.7.6 4.6 The Simulation Gallery

Twelve instances are accompanied by standalone interactive HTML simulations accessible at <https://echopraxium.github.io/tscg/>. These simulations are **pedagogical instruments** — not engineering simulations in the quantitative sense.

Two rendering technologies are used:

- **BabylonJS 6.26.0** (3D): Primary library for immersive spatial simulations.
- **p5.js** (Canvas2D): For systems whose dynamics are better represented in 2D.
- **Tone.js**: Audio simulations (Counterpoint).

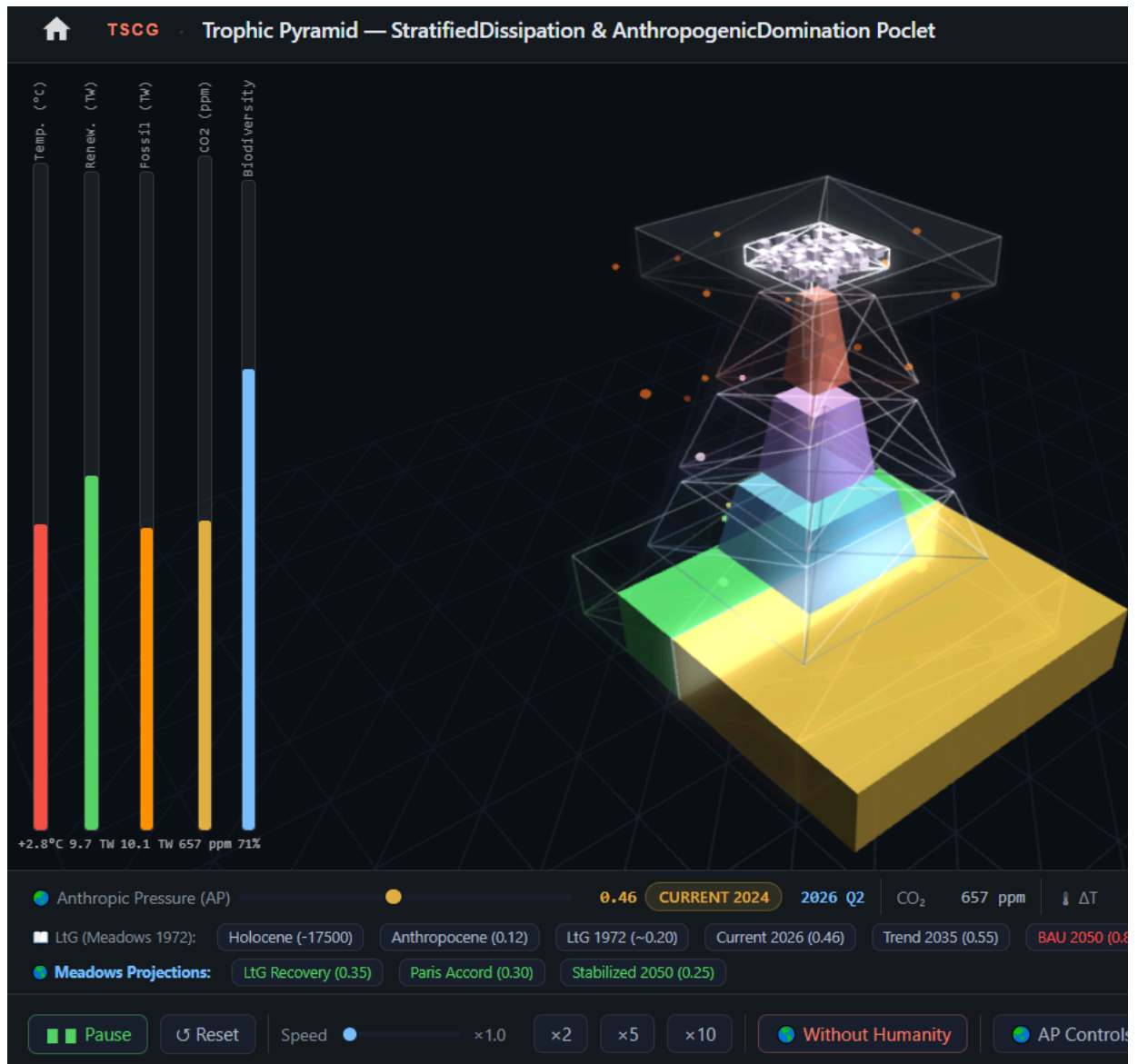


Figure 5 — TrophicPyramid Poclet BabylonJS simulation

Figure 5 — The TrophicPyramid Poclet BabylonJS 3D simulation (Ecology/Biology, v2.1.0). Six trophic levels (Producers, Decomposers, Herbivores, Carnivores, Apex Predators, Humanity) are rendered as a layered pyramid with Lindeman's 10% energy transfer rule ($\eta \approx 0.10$). The simulation integrates an Anthropogenic Pressure (AP) model grounded in Meadows et al. (1972), with real-time display of CO₂ (657 ppm), temperature (+2.8°C), biodiversity index (71%), and Meadows projections (LtG 1972, Paris Accord, BAU 2050). Humanity is modeled as a supra-trophic super-predator — a GenericConcept extension of the standard trophic hierarchy.

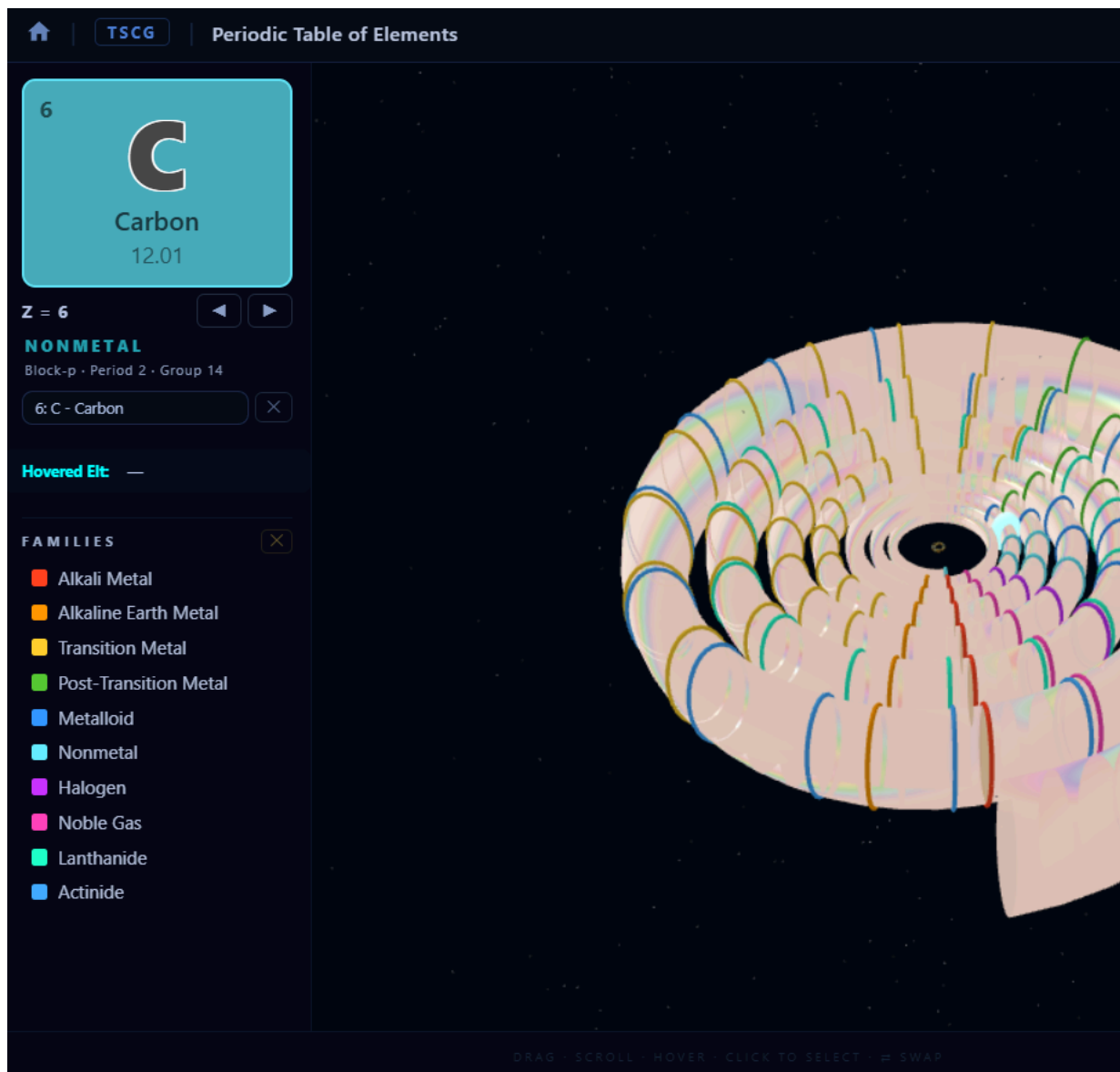


Figure 6 — Ptoe Poclet BabylonJS simulation

Figure 6 — The Periodic Table of Elements (Ptoe) Poclet BabylonJS 3D simulation. 118 elements are arranged as a navigable spiral structure, color-coded by family (Alkali Metals, Transition Metals, Halogens, Noble Gases, etc.). The simulation demonstrates how quantum constraints (orbital structure, valence electrons) generate emergent periodic patterns — a TSCG Hierarchy $\times$ Symmetry $\times$ Invariant system. The interactive interface displays element cards (atomic number, mass, block, period, group) alongside family descriptions.

1.7.7 4.7 SHACL Validation Infrastructure

Every instance is validated against `M0_Instances_Schema.shacl.ttl`, which enforces 9 mandatory constraints including presence of `m3:ontologyType`, ASFID/REVOI score completeness, `dcterms:creator` format, changelog rolling maximum of 3 entries, and `SpectralClass/FocalClass` drawn from OWL enumeration IRIs.

1.8 5. Epistemic Scoring

1.8.1 5.1 Overview: Two Complementary Metrics

TSCG's epistemic scoring system quantifies the relationship between Territory measurement and Map construction for any instance. Two complementary metrics are defined:

- δ_1 (Epistemic Gap): measures the *distance* between ASFID and REVOI score vectors — how well the Map tracks the Territory.
- δ_2 / EFS (Epistemic Focal Score): measures the *quality of stereopsis fusion* — how deeply and equitably the Gs/TKSL layer triangulates the two perspectives.

All scores are stored as flat numeric values at the `owl:Ontology` root level of the M0 JSON-LD file:

```
"m0:scoreA": 0.85, "m0:scoreS": 0.90, "m0:scoreF": 0.80,
"m0:scoreIt": 0.75, "m0:scoreD": 0.70,
"m0:scoreR": 0.85, "m0:scoreE": 0.75, "m0:scoreV": 0.70,
"m0:scoreO": 0.80, "m0:scoreIm": 0.65
```

1.8.2 5.2 δ_1 — The Epistemic Gap

ASFID_mean = (scoreA + scoreS + scoreF + scoreIt + scoreD) / 5
REVOI_mean = (scoreR + scoreE + scoreV + scoreO + scoreIm) / 5
 δ_1 = |ASFID_mean - REVOI_mean| / $\sqrt{2}$

δ_1 is classified into four **SpectralClasses**:

SpectralClass	δ_1 range	Interpretation
Coherent	[0, 0.05)	Near-perfect alignment
OnCriticalLine	[0.05, 0.15)	Productive tension
Liminal	[0.15, 0.30)	Significant gap
Enigmatic	[0.30, 1.0)	Fundamental challenge

1.8.3 5.3 δ_2 / EFS — The Epistemic Focal Score

stereopsisDepth = (Tx0.25) + (Kx0.25) + (Ssx0.25) + (Lx0.25)
focalBias = REVOI_mean - ASFID_mean $\in [-1, +1]$
 δ_2 / EFS = stereopsisDepth $\times (1 - |focalBias|)$ $\in [0, 1]$

focalBias indicates the directional lean of the instance: - `focalBias > 0` → Map-dominant (Hyperopic) - `focalBias < 0` → Territory-dominant (Myopic) - `focalBias = 0` → perfectly balanced (Emmetropic candidate)

δ_2 is classified into six **FocalClasses**:

FocalClass	EFS range	focalBias	Interpretation
Emmetropic	≥ 0.70	any	Deep, balanced fusion
SlightlyMyopic	[0.55, 0.70)	< 0	Slight Territory bias
SlightlyHyperopic	[0.55, 0.70)	> 0	Slight Map bias
Myopic	[0.35, 0.55)	< 0	Territory dominant
Hyperopic	[0.35, 0.55)	> 0	Map dominant
Astigmatic	< 0.35	any	Collapsed fusion

Conditional: EFS is only computed when `m0:focalApplicable = true` (at least one stereopsis GenericConcept mobilized by the instance).

1.8.4 5.4 The scoringStatus Property

scoringStatus	Meaning
<code>m0:scoringStatus.Complete</code>	All 10 ASFID+REVOI scores present
<code>m0:scoringStatus.Partial</code>	Some scores assigned
<code>m0:scoringStatus.Pending</code>	No scores — stub instance

`M0_AdaptativeImmuneResponse.jsonld` is the reference implementation for the complete scoring convention.

1.8.5 5.5 Scoring as a Research Instrument

The scoring system is not a quality ranking — it is a **research instrument** for identifying modeling gaps, comparing structural homologs across domains, and tracking framework maturation. A high δ_1 (Enigmatic) points to specific dimensions where the model is weak; two instances with similar ASFID/REVOI profiles are candidates for structural homology across disciplines.

1.9 6. Validation

1.9.1 6.1 Epistemological Stance: Falsificationism, Not Proof

TSCG adopts an explicitly **falsificationist stance** (Popper, 1959): the framework is submitted as a conjecture that has not yet been invalidated, not as a theory that has been confirmed. The corpus of 33 validated instances constitutes the current empirical record.

A decisive counter-example would take one of three forms:

1. **Dimensional collapse:** a well-understood system that demonstrably cannot be characterized by any combination of ASFID and REVOI dimensions.
2. **Concept redundancy:** a demonstration that two GenericConcepts are structurally identical (same formula, same domain coverage).
3. **Purity violation:** a demonstration that a claimed M2 GenericConcept is valid in fewer than six unrelated knowledge fields.

The research community is explicitly invited to attempt these falsifications.

1.9.2 6.2 The Six-Domain Threshold

The primary guard against Ontological Overfitting: a GenericConcept earns M2 membership only by demonstrating structural relevance across at least six unrelated knowledge fields.

Representative validations:

- Dissipation** (F × D): thermodynamics, electronics, biology, economics, distributed systems, music. Six domains → M2 confirmed.
- Homeostasis** (A × St × F | L): biology, economics, electronics, ecology, distributed systems, mythology. Six domains → confirmed.
- Coherence** (A × St × It | R + O | _^/_\$): physics, distributed systems, music, biology, linguistics, photography. Six domains → confirmed.

1.9.3 6.3 The Corpus as Empirical Record

The 33-instance corpus spans twelve knowledge domains and four instance types. Three structural observations:

- Cross-domain SpectralClass consistency:** instances from unrelated domains tend to cluster in similar SpectralClasses when they share dominant GenericConcepts — consistent with the transdisciplinary homology hypothesis, though not yet statistically validated.
- Energetic family universality:** every instance in the corpus mobilizes at least one concept from the Energetic family (Dissipation, Entropy, Transducer, Processor, or Storage). This near-universality was not designed — it emerged from independent instance analyses.
- Gs/TKSL activation:** EFS scoring (δ_z) has not yet been computed for the current corpus — m0:focalApplicable is absent in all 33 instances as of v16.2.0. This constitutes a known gap in the scoring infrastructure, documented in PENDING_SCORES_README.md . EFS computation is identified as a priority for the next release cycle.

1.9.4 6.4 The TriskeleToolchain Validation

The most architecturally significant validation in the corpus is the **TriskeleToolchain SymbolicSystemGrammar** — an independent structural validation of the Base16 from virtual machine design and compiler engineering.

The TriskeleToolchain ISA comprises **256 opcodes organized into exactly 16 categories**, one per M3 primitive, mapped bijectively to the Base16:

Gt (Territory ×)	0x0_ A_	0x1_ St_	0x2_ F_	0x3_ It_	0x4_ D_
Gm (Map +)	0x5_ R_	0x6_ E_	0x7_ V_	0x8_ O_	0x9_ Im_
Gs (Stereopsis)	0xA_ T_	0xB_ _^_	0xC_ _\$ _	0xD_ K_	0xE_ Ss_ 0xF_ L_

Each opcode category is **semantically deductible** from its M3 primitive:

- F_ (Flow): jump, call, return, yield — “How does execution flow?”
- A_ (Attractor): push, pop, alloc, free — “Where do data converge?”
- K_ (Knowledge): typeof, is_a, schema — “What is known at runtime?”
- _^_ / _\$ _ : 14 exact symmetric pairs with no residue.

Four properties make this validation particularly strong: **sufficiency** (256 opcodes cover Wolf3D + Doom without additional categories), **non-redundancy** (no overlap between categories after semantic stress-testing), **cross-validation of K, Ss, L** (the three Gs primitives added to resolve 17 M2 formula collisions are exactly those completing the ISA to Base16 — double independent validation), and **^/\$ symmetry** (14 exact pairs confirming structural completeness of polar modifiers).

1.9.5 6.5 Ontological Overfitting: The Central Anti-Pattern

Ontological Overfitting — adding a GenericConcept to M2 because it appears prominently in one well-studied system — is the most persistent threat to M2 integrity. It manifests in three patterns:

- The prominence trap:** “membrane” in biology → decomposes into Constraint × Gradient × Selectivity → belongs in M1_Biology, not M2.
- The naming trap:** “Apoptosis” sounds biology-specific; “Absorbing State” captures the same pattern across biology, economics, physics, and distributed systems.
- The granularity trap:** “Negative feedback” is a specialization of Regulation (feedback_polarity attribute) — an attribute value, not a separate M2 concept.

1.9.6 6.6 Limitations and Open Questions

- Score subjectivity:** ASFID/REVOI scores are assigned through structured expert judgment. A Multisubjective Score Evaluation Protocol is under development.
- Mathematical formalism pending expert review:** the Structural Grammar foundation was developed with Claude AI and has not been reviewed by a category theory expert.
- Corpus size:** 33 instances across twelve domains is a promising but modest empirical base.
- Observer dependence of REVOI:** Map scores are inherently observer-relative.
- Simulation ≠ prediction:** the interactive simulations are pedagogical instruments, not quantitative predictive models.

1.10 7. Discussion

1.10.1 7.1 What TSCG Is — and Is Not

TSCG is not a Theory of Everything. It is a **construction kit** evaluated by its expressiveness (33-instance corpus across twelve domains), economy (80 GenericConcepts from 16 primitives), and productivity (structural homologies between biological filtration and distributed consensus, between mythological cosmology and network topology, between blockchain proof-of-work and thermodynamic dissipation).

What TSCG does not claim: completeness, uniqueness, or optimality. The 80 GenericConcepts are *a* vocabulary that has survived internal scrutiny. Alternative decompositions may exist and may prove superior.

1.10.2 7.2 The Structural Grammar Migration: A Foundational Step

The migration from Hilbert-space approximation to Structural Grammar (Lambek calculus, free commutative monoidal categories) in v16.x represents the most significant foundational change in TSCG's history.

The earlier formalism used tensor product notation and Hilbert space metaphors — a pragmatic choice that enabled rapid formalization but carried an implicit claim (metric structure, inner product, orthogonality) that TSCG's concepts do not satisfy and do not need.

The Structural Grammar migration replaces this with a more honest formalism: the monoidal product operators require no metric, no inner product, and no notion of orthogonality. Homeostasis is not a vector in a Hilbert space — it is a monoidal expression $A \times St \times F \mid L$ whose meaning emerges from the joint activation of its primitives.

This migration: removes an unwarranted implicit claim; provides a legitimate algebraic foundation; and opens a path toward formal machine verification via SHACL shapes.

1.10.3 7.3 Two Named Failure Modes: Narcissus and Icarus

TSCG names two archetypal failure modes that any systemic modeling framework must guard against.

Narcissus — the self-referential trap. A modeler mistakes the model for the territory: the Map becomes so absorbing that the modeler forgets it is a representation, not reality. In TSCG terms, Narcissus produces instances with very high REVOI scores but systematically low ASFID scores — δ_1 grows large (Enigmatic) while the modeler, absorbed in the elegance of the Map, fails to notice.

The defense: mandatory ASFID scoring forces the modeler to return to empirical Territory observation. The Map is always the map of *something*.

Icarus — the overreach trap. A modeler attempts to model everything with the same vocabulary, regardless of whether the system warrants it. In TSCG terms, Icarus produces Ontological Overfitting: new GenericConcepts added to M2 for every new system encountered, until the vocabulary collapses under its own weight.

The defense: the six-domain threshold and the M2 Purity Principle force the modeler to pause and ask: is this concept truly irreducible, or am I just excited by a new system?

Together, Narcissus and Icarus define the two edges of the modeling corridor that TSCG attempts to navigate: not so absorbed in the Map that the Territory is forgotten; not so ambitious in the vocabulary that generality is sacrificed.

1.10.4 7.4 TSCG and Prior Art: Similarities and Distinctions

General System Theory (von Bertalanffy, 1968): GST identified the *existence* of cross-domain structural patterns; TSCG attempts to *enumerate and formalize* them as a combinatorial vocabulary with monoidal structure.

Cybernetics (Wiener, 1948; Ashby, 1956): TSCG's ASFID dimensions are deeply cybernetic in spirit. TSCG extends scope to non-regulatory phenomena (Entropy, Emergence, Bifurcation) outside classical cybernetic scope.

VSM (Beer, 1972): A concrete precursor to the desiloification hypothesis. TSCG re-encodes VSM as a SystemicFramework, validating structural homology.

TRIZ (Altshuller, 1946–1985): The earliest large-scale empirical validation of the desiloification hypothesis known to the author. TSCG re-encodes TRIZ as a SystemicFramework.

Category Theory (Mac Lane, 1971): TSCG is not a contribution to category theory — it is an *application* of categorical structures to ontology engineering.

Where TSCG claims genuine novelty — pending expert review — is in the *combination*: a combinatorial vocabulary of atomic transdisciplinary patterns organized as monoidal products of a small primitive set (Base16), validated through a growing instance corpus spanning radically heterogeneous domains, with a two-dimensional epistemic scoring system (δ_1/δ_2) measuring alignment between Territory observation and Map construction.

1.10.5 7.5 The Human-AI Collaboration as a Research Method

TSCG's development has been inseparable from its collaboration with Claude AI — methodologically significant and deserving explicit reflection, including an honest account of both its productivity and its limitations.

1.10.5.1 The Indispensable Role of Human Steering

Human steering is not optional: it is structurally necessary.

Claude AI optimizes locally: toward well-formed outputs, toward completeness, toward internally consistent artifacts. Left unsteered, the collaboration would produce ontologically overfit GenericConcepts, prematurely stabilized formalisms, and technically correct but semantically hollow artifacts.

Two concrete examples illustrate the necessity of active human discernment:

Ontology property naming (M0 layer): During early instance development, property naming conventions across M0 ontologies drifted silently — different instances used inconsistent property names for structurally identical concepts. This inconsistency was not caught by Claude AI, which optimized each instance in isolation. The corrective required *a posteriori* extraction of a SHACL grammar from the M0 corpus, followed by systematic realignment of all existing instances against the extracted schema. The SHACL grammar (`M0_Instances_Schema.shac1.tt1`) now serves as the machine-verifiable guard against future drift — but its existence was motivated by a failure that human oversight had to diagnose.

BabylonJS 3D simulation (local vs global reference frames): During development of a 3D instance simulation, a persistent spatial inconsistency arose from a confusion between object-local and scene-global reference frames. Claude AI, generating code without prior embodied experience of the library, could not identify the source of the impasse. The author — having prior hands-on experience with BabylonJS before AI-assisted development — recognized the reference frame issue immediately and resolved it. This illustrates a broader principle: **AI-generated code requires a human capable of validating it** — not merely reviewing it syntactically, but understanding its behavior in context.

1.10.5.2 What the Collaboration Delivers

Within the bounds of active human steering and discernment, the Human-AI collaboration has been genuinely transformative. Claude AI has been extremely useful across four domains:

- **Bibliographic research:** rapid, targeted literature synthesis for each new instance domain, producing domain-specific context that would have required weeks of manual research per instance.
- **Code generation:** standalone HTML simulations (BabylonJS 3D, p5.js Canvas2D), Python migration scripts, SHACL validators, and JavaScript ontology loaders.
- **Ontology engineering:** JSON-LD encoding of M0 instances, M1 domain extensions, and M2/M3 ontology updates — respecting W3C standards and TSCG namespace conventions — delegated to Claude AI with human review at each step.
- **Technical documentation:** READMEs, changelog entries, SKILL pipeline specifications, and this article — drafted by Claude AI from the author’s conceptual specifications, revised iteratively under the author’s editorial direction.

1.10.5.3 A Methodological Hypothesis

This experience suggests that **long-horizon conceptual research** — requiring sustained intuition over years combined with intensive formalization effort — may be a domain where Human-AI collaboration produces results that neither party could achieve independently, provided the human maintains active steering capacity throughout.

The key condition: the human must be capable not only of specifying what is wanted, but of **evaluating what is produced** — detecting semantic drift in ontologies, diagnosing behavioral failures in generated code, and resisting the local optimization pressures that generative AI systems naturally exert.

1.10.6 7.6 On Being a Work in Gestation

TSCG is submitted to the research community as a **work in active gestation** — not a finished product, not a validated theory, but a structured invitation.

The honest assessment of its current state:

- The **mathematical foundation** is more rigorous than its predecessor (Structural Grammar vs Hilbert-space approximation) but has not been reviewed by a category theory expert.
- The **instance corpus** is growing but modest: 33 instances across twelve domains, with EFS scoring not yet applied to any instance.
- The **scoring system** is principled but subjective: inter-analyst variance has not been quantified, and the Multisubjective Score Evaluation Protocol is under development.
- The **simulation gallery** is pedagogically valuable but not predictively rigorous.

What TSCG offers at this stage is not proof but **provocation**: a structured vocabulary for asking whether structural homologies exist across disciplinary silos, and a growing corpus of modeling attempts that have survived internal scrutiny without producing a decisive counter-example.

The framework is submitted not imposed. It is a kitchen, not a recipe. The author is looking not for customers but for **cooks** — researchers and practitioners willing to test whether this arbitrary grid can be broken, extended, or redirected toward problems it was not designed to address.

If TSCG’s GenericConcepts are wrong, the research community will find the counter-examples. If they are right — or right enough — the desiloification hypothesis will have found its first operational vocabulary.

Either outcome advances knowledge.

1.11 8. References

1.11.1 Foundational Theory

Ashby, W. R. (1956). *An Introduction to Cybernetics*. Chapman & Hall.

Berners-Lee, T., Hendler, J., & Lassila, O. (2001). The Semantic Web. *Scientific American*, 284(5), 34–43.

Bertalanffy, L. von (1968). *General System Theory: Foundations, Development, Applications*. George Braziller.

Beer, S. (1972). *Brain of the Firm*. Allen Lane.

Eilenberg, S., & Mac Lane, S. (1945). General theory of natural equivalences. *Transactions of the American Mathematical Society*, 58(2), 231–294.

Korzybski, A. (1933). *Science and Sanity: An Introduction to Non-Aristotelian Systems and General Semantics*. Institute of General Semantics.

Lambek, J. (1958). The mathematics of sentence structure. *American Mathematical Monthly*, 65(3), 154–170.

Lambek, J. (1988). Categorical and categorical grammars. In R. T. Oehle, E. Bach, & D. Wheeler (Eds.), *Categorical Grammars and Natural Language Structures* (pp. 297–317). Springer.

Mac Lane, S. (1971). *Categories for the Working Mathematician*. Springer.

Maturana, H. R., & Varela, F. J. (1980). *Autopoiesis and Cognition: The Realization of the Living*. D. Reidel.

Peirce, C. S. (1931–1958). *Collected Papers* (Vols. 1–8). Harvard University Press.

Popper, K. R. (1959). *The Logic of Scientific Discovery*. Hutchinson.

Saussure, F. de (1916). *Cours de linguistique générale*. Payot.

Wiener, N. (1948). *Cybernetics: Or Control and Communication in the Animal and the Machine*. MIT Press.

1.11.2 Systemic Frameworks

Altshuller, G. S. (1996). *And Suddenly the Inventor Appeared: TRIZ, the Creative Problem Solving*. Technical Innovation Center.

Osterwalder, A., & Pigneur, Y. (2010). *Business Model Generation*. John Wiley & Sons.

1.11.3 Ontology Engineering & Semantic Web

Brickley, D., & Guha, R. V. (2014). *RDF Schema 1.1*. W3C Recommendation. <https://www.w3.org/TR/rdf-schema/>

Hitzler, P., et al. (2012). *OWL 2 Web Ontology Language Primer* (2nd ed.). W3C Recommendation. <https://www.w3.org/TR/owl2-primer/>

Knublauch, H., & Kontokostas, D. (2017). *Shapes Constraint Language (SHACL)*. W3C Recommendation. <https://www.w3.org/TR/shacl/>

1.11.4 Mathematical & Algebraic Foundations

Coecke, B., & Paquette, É. O. (2011). Categories for the practising physicist. In B. Coecke (Ed.), *New Structures for Physics* (pp. 173–286). Springer.

Fong, B., & Spivak, D. I. (2019). *An Invitation to Applied Category Theory: Seven Sketches in Compositionality*. Cambridge University Press.

1.11.5 Human-AI Collaboration

Anthropic. (2024). *Claude: A large language model*. Anthropic PBC. <https://www.anthropic.com>

1.11.6 TSCG Primary Sources

Kern, M. (aka Echopraxium), with the collaboration of Claude AI. (2026). *TSCG: Transdisciplinary System Construction Game — A Systemic Modeling Toolkit* [Prior version v5.0]. Zenodo. <https://doi.org/10.5281/zenodo.1954443>

Kern, M. (aka Echopraxium), with the collaboration of Claude AI. (2026). *TSCG GitHub Repository* (v16.2.0). <https://github.com/Echopraxium/tscg>

Kern, M. (aka Echopraxium), with the collaboration of Claude AI. (2026). *TSCG Simulation Gallery*. <https://echopraxium.github.io/tscg/>

1.12 Appendix A — Complete Instance Corpus (v16.2.0)

33 validated instances across 4 types. All files at: <https://github.com/Echopraxium/tscg/tree/main/instances/>

1.12.1 A.1 Pociets (25)

Instance	Domain	Simulation
AdaptativeImmuneResponse	Biology	—
BloodPressureControl	Biology	—
ButterflyMetamorphosis	Biology	—
CellSignalingModes	Biology	—
ColorSynthesis (<i>féderateur</i>)	Optics	—
ComplexChemicalSynapse	Biology	—
Counterpoint	Music	Tone.js
ExposureTriangle	Photography	BabylonJS
FireTriangle	Chemistry	p5.js
FourStrokeEngine	Mechanical Eng.	BabylonJS
Kidneys	Biology	—
KindlebergerMinsky	Economics	BabylonJS
MtgColorWheel	Gaming	BabylonJS
NakamotoConsensus	Distributed Systems	BabylonJS
NuclearReactorsTypology	Nuclear Eng.	BabylonJS
PhaseTransition	Physics	BabylonJS
PlateTectonics	Geology	BabylonJS
Ptoe	Physics / Chemistry	BabylonJS
Raas	Biology	—
Tpack	Pedagogy	BabylonJS
Transistor	Electronics	BabylonJS
TrophicPyramid	Ecology	—
TvTestPattern	Signal Processing	p5.js
Vco	Electronics	BabylonJS
Yggdrasil	Norse Mythology	BabylonJS

1.12.2 A.2 SystemicFrameworks (3)

Instance	Origin	Files	Simulation
VSM	Beer (1972)	M0_Vsm.jsonld	—
TRIZ	Altshuller (1946–1985)	M0_Triz.jsonld	—
BMC	Osterwalder (2004)	M0_Bmc.jsonld + M0_BmcSimulation.jsonld	p5.js

1.12.3 A.3 SymbolicSystemGrammars (2)

Instance	Domain	File	Note
IChing	Divination / Combinatorics	M0_IChing.jsonld	64 hexagrams as binary combinatorial system
TriskeleToolchain	VM Design / Compiler Eng.	M0_TriskeleToolchain.jsonld	ISA Base16 — 256 opcodes across 16 M3 categories

1.12.4 A.4 TscgTools (3)

Instance	Technology	Role
TscgOntologyExplorer	ElectronJS	Desktop ontology navigator
TscgOntologyAPIServer	Node.js REST	Local API server for ontology access
TscgPocletMiner	ElectronJS + LLM	AI-assisted instance generation

1.13 Appendix B — Human-AI Collaboration Methodology

1.13.1 B.1 Infrastructure

Component	Role
Claude Projects	Persistent corpus of reference documents across sessions
Smart Prompt	Structured document summarizing architectural state — re-injected at each session
Skills pipelines	Reusable procedural specifications encoding the author’s methodology

1.13.2 B.2 Active Skills

Skill	Purpose
tscg-instance-pipeline	4-step instance creation: Proposition → Analysis → Modeling → Simulation
tscg-article-pipeline	5-step article pipeline: Audit → Planning → Drafting → Revision → Finalization
tscg-create-instance-simulation	BabylonJS 3D simulation generation
tscg-ontology-diagnosis-pipeline	Safe ontology engineering with 6-phase validation
tscg-tensor-to-structural-grammar-migration	Migration from Hilbert/tensor to Lambek/monoidal formalism
tscg-generate-mn-grammars	SHACL grammar generation for M1/M2/M3
tscg-triskelevm-debug	TriskeleToolchain diagnostic and debugging
tscg-rust-vm	TriskeleVM Rust implementation

1.13.3 B.3 Task Delegation Matrix

Task	Human	Claude AI
Systemic intuition & pattern recognition	✔ Primary	—
Architectural discipline & Overfitting rejection	✔ Primary	Advisory
Domain knowledge validation	✔ Primary	—
Artifact evaluation & discernment	✔ Required	—
Bibliographic research	Advisory	✔ Primary
Monoidal formula derivation	Specification	✔ Primary
JSON-LD ontology encoding	Review & validation	✔ Primary
SHACL constraint generation	Review & validation	✔ Primary
Code generation (JS, Python, Rust)	Steering & validation	✔ Primary
Technical documentation	Direction & editing	✔ Primary
GenericConcept candidate evaluation	✔ Primary	Proposal
Instance simulation design	Direction & ergonomics	✔ Primary

1.14 Appendix C — M2 GenericConcepts Catalog (v16.14.0)

80 atomic GenericConcepts across 9 families. Formulas use monoidal operators: $\times$ (Territory/Gt), $+$ (Map/Gm), $\sqcup$ (Stereopsis/Gs). Priority: $\times$ > $+$ > $\sqcup$. Full ontology: https://github.com/Echopraxium/tscg/blob/main/ontology/M2_GenericConcepts.jsonld

1.14.1 C.1 Structural Family (23 concepts)

Concept	Formula	Polarity
Capacity	$St \times It \setminus L$	neutral
Channel	$St \times F \setminus Ss$	neutral
Cluster	$S \times I \times A$	neutral
Coherence	$A \times St \times It \setminus R + O \setminus _{}^{\wedge} / _{}^{\$}$	dual
Component	$St \times It \setminus L$	neutral
Composition	$St \times It \times A \setminus _{}^{\wedge} / _{}^{\$}$	dual
Hierarchy	$S \times A$	neutral
Hub	$S \times F$	neutral
Identity	$St \times It \times A \setminus V + E$	dual
Interoperability	$St \times It \times F \setminus V + E$	neutral
Invariant	$S \times A$	neutral
Layer	$St \times It \times A \setminus R$	neutral
Modularity	$S \times I$	neutral
Network	$S \times I \times F \setminus O + R + Im + V$	neutral
Node	$St \times It \setminus L$	neutral
Path	$S \times I \times D$	neutral
Polarity	$S \times I \times A$	n-ary
Segmentation	$S \times I \times D$	neutral
Stase	$S \times A$	neutral
Step	$S \times I \times D$	dual
Symmetry	S	neutral
Topology	$St \setminus L$	neutral
Workflow	$S \times D \times F$	dual

1.14.2 C.2 Dynamic Family (17 concepts)

Concept	Formula	Polarity
Action	$D \times I$	dual
Alignment	$I \times A \times S$	neutral
Amplification	$F \times D \times It \setminus R + O \setminus _{}^{\wedge} / _{}^{\$}$	hybrid
Behavior	$S \times D \times F$	dual
Bifurcation	$D \times F \setminus L$	neutral
Convergence	$F \times D \setminus _{}^{\wedge} / _{}^{\$}$	dual
Duplication	$St \times It \times F \times D \setminus V$	neutral
Event	$D \times I$	neutral
FeedbackLoop	$Process \times Alignment \times Homeostasis$	dual
Fusion	$St \times D \setminus _{}^{\wedge} / _{}^{\$}$	dual
Modelisation	$D \times F \times It \setminus R + V + E$	neutral
Process	$D \times F$	neutral
Processor	$S \times I \times F \times D$	neutral
Synergy	$It \times D \setminus _{}^{\wedge} / _{}^{\$}$	dual
Trajectory	$A \times D \times F$	neutral
Transformation	$D \times S \times I$	neutral
Tropism	$A \times S \times D \times F$	dual

1.14.3 C.3 Ontological Family (10 concepts)

Concept	Formula	Polarity
Context	$O + R + Im + E$	neutral
Environment	$F \times I$	neutral
Gradient	$F \times T$ or $I \times T$	neutral
Imbrication	$S \times S$	dual
Observer	$I \times A$	neutral
Resource	$F \times It \setminus \setminus R$	neutral
Space	$St \setminus \setminus L$	neutral
State	I	neutral
Substrate	$S \times F$	neutral
System	$S \times F$	neutral

1.14.4 C.4 Regulatory Family (10 concepts)

Concept	Formula	Polarity
Activation	$A \times D \setminus \setminus _{}^{\wedge} / _{}^{\$}$	dual
Balance	$A \times S \times F \setminus \setminus _{}^{\theta}$	dual
Constraint	$St \setminus \setminus L$	neutral
Homeostasis	$A \times St \times F \setminus \setminus L$	neutral
Regulation	$A \times S \times F \setminus \setminus R + V + E$	neutral
Scope	$St \times It \times A \setminus \setminus R + V$	neutral
Threshold	$A \times I$	neutral
Trade-off	$A \times I \times F$	neutral
TriadicBalance	$A \times F \times D \setminus \setminus _{}^{\theta}$	ternary
Trigger	$D \times I$	neutral

1.14.5 C.5 Informational Family (8 concepts)

Concept	Formula	Polarity
Code	$It \times Ss$	neutral
Coding	$It \times St \times D \setminus \setminus _{}^{\wedge} / _{}^{\$}$	dual
Language	$Ss \times F \setminus \setminus K$	neutral
Pattern	$St \times It \times A \setminus \setminus K \setminus \setminus _{}^{\wedge} / _{}^{\$}$	dual
Representation	$I \times S$	neutral
Signal	$I \times F$	neutral
Signature	$It \times Ss \setminus \setminus V$	neutral
ValueSpace	$It \setminus \setminus V + O + R + Im$	hybrid

1.14.6 C.6 Adaptive Family (4 concepts)

Concept	Formula	Polarity
Adaptation	$I \times F \times D$	neutral
Emergence	$I \times S \times D$	neutral
Memory	$\int (D-F) d\tau$	neutral
Resilience	$A \times S$	neutral

1.14.7 C.7 Energetic Family (4 concepts)

Concept	Formula	Polarity
Dissipation	$F \times D$	neutral
Entropy	$F \times It \times D \setminus \setminus _{}^{\wedge} / _{}^{\$}$	dual
Storage	$S \times F$	neutral
Transducer	$F \times S \times I$	neutral

1.14.8 C.8 Relational Family (5 concepts)

Concept	Formula	Polarity
Agent	$St \times It \times D \setminus K$	neutral
Link	$S \times I \times F$	neutral
Mediator	$F \times It \setminus K$	neutral
Relation	$St \times Ss$	neutral
Role	$Ss \setminus K$	neutral

1.14.9 C.9 Teleonomic Family (1 concept)

Concept	Formula	Polarity
Self-Organization	$A \times I \times D$	neutral

1.14.10 C.10 Combo Family (4 validated GenericConceptCombos)

Combo	Formula	Components	Polarity
AbsorbingState	$S \times A \times F \times I \times D$	$\Rightarrow$ (Stase, Entropy)	neutral
Inertia	$S \times F \times I \times D$	$\Rightarrow$ (Memory, Entropy)	neutral
Potentialization	$A \times D \times F \setminus _ \wedge / _ \$$	$\Rightarrow$ (Activation, Process)	dual
TopologicalDefect	$St \times A \times It \setminus R + O$	$\Rightarrow$ (Incoherence, Invariant)	neutral

1.15 Appendix D — Scoring Reference

1.15.1 D.1 ASFID/REVOI Score Properties

Property	Dimension	Monoid
$m\theta:scoreA$	Attractor	Gt
$m\theta:scoreS$	Structure	Gt
$m\theta:scoreF$	Flow	Gt
$m\theta:scoreIt$	Information (Territory)	Gt
$m\theta:scoreD$	Dynamics	Gt
$m\theta:scoreR$	Representability	Gm
$m\theta:scoreE$	Evolvability	Gm
$m\theta:scoreV$	Verifiability	Gm
$m\theta:scoreO$	Observability	Gm
$m\theta:scoreIm$	Interoperability (Map)	Gm

 $m\theta:scoreIt \neq m\theta:scoreIm$ — systematic disambiguation required.

1.15.2 D.2 Derived Metrics

$$\begin{aligned} ASFID_mean &= (scoreA + scoreS + scoreF + scoreIt + scoreD) / 5 \\ REVOI_mean &= (scoreR + scoreE + scoreV + scoreO + scoreIm) / 5 \\ \delta_1 &= |ASFID_mean - REVOI_mean| / \sqrt{2} \end{aligned}$$

1.15.3 D.3 SpectralClass Enumeration

IRI	δ_1 Range	Meaning
$m\theta:spectralClass.Coherent$	[0, 0.05)	Near-perfect alignment
$m\theta:spectralClass.OnCriticalLine$	[0.05, 0.15)	Productive tension
$m\theta:spectralClass.Liminal$	[0.15, 0.30)	Significant gap
$m\theta:spectralClass.Enigmatic$	[0.30, 1.0)	Fundamental challenge

1.15.4 D.4 EFS — Epistemic Focal Score (δ_2)

$$\begin{aligned} stereopsicDepth &= (T \times 0.25) + (K \times 0.25) + (Ss \times 0.25) + (L \times 0.25) \\ focalBias &= REVOI_mean - ASFID_mean \\ \delta_2 / EFS &= stereopsicDepth \times (1 - |focalBias|) \end{aligned}$$

Conditional: only when $m\theta:focalApplicable = true$.

1.15.5 D.5 FocalClass Enumeration

IRI	EFS Range	focalBias	Meaning
<code>m0:focalClass.Emmetropic</code>	≥ 0.70	any	Sharp combined focus
<code>m0:focalClass.SlightlyMyopic</code>	[0.55, 0.70)	< 0	Slight Territory bias
<code>m0:focalClass.SlightlyHyperopic</code>	[0.55, 0.70)	> 0	Slight Map bias
<code>m0:focalClass.Myopic</code>	[0.35, 0.55)	< 0	Territory dominant
<code>m0:focalClass.Hyperopic</code>	[0.35, 0.55)	> 0	Map dominant
<code>m0:focalClass.Astigmatic</code>	< 0.35	any	No stereopsis possible

1.15.6 D.6 ScoringStatus Enumeration

IRI	Meaning
<code>m0:scoringStatus.Complete</code>	All 10 ASFID+REVOI scores present
<code>m0:scoringStatus.Partial</code>	Some scores present
<code>m0:scoringStatus.Pending</code>	No scores — stub instance

1.15.7 D.7 Reference Implementation

`M0_AdaptativeImmuneResponse.jsonld` is the canonical reference for a fully scored instance with both δ_1 and δ_2 computed.

1.16 Appendix E — Framework Maturity Assessment (v16.2.0)

Dimension	Status	Notes
M3 GenesisGrammar	🟢 Stable	Structural Grammar migration complete (v16.x)
M2 GenericConcepts	🔄 Active	80 concepts, 9 families; open to additions via six-domain threshold
M1 Domain Extensions	🔄 Active	13 domains; new extensions added per instance
M0 Instance Corpus	🔄 Growing	33 validated instances; EFS scoring pending
SHACL Grammar	🟢 Stable	v1.5 — 9 mandatory constraints
Mathematical foundation	⚠️ Pending	Lambek/monoidal — expert review required
EFS scoring (δ_2)	⚠️ Pending	<code>focalApplicable</code> not yet applied to corpus
Multisubjective Protocol	🔧 In dev	Inter-analyst variance quantification
Simulation Gallery	🟢 Live	12 simulations at echopraxium.github.io/tscg
TriskeleToolchain	🔄 Active	v0.3.8 — Wolf3D + Doom partial validation
Zenodo publication	🟢 Published	DOI: 10.5281/zenodo.19544443 (v5.0); v6.0 pending

1.17 Appendix F — Core Hypotheses & Theoretical Foundations

Documents available in the repository under `ontology/docs/` :

1.17.1 F.1 Core Hypotheses

#	Document	Core Claim
H1	<code>TerritoryMap_Dichotomy.md</code>	The Map-Territory distinction (Korzybski) extended as a formalized bidirectional feedback loop — the epistemological foundation of the bicephalous architecture
H2	<code>SystemicEsperanto.md</code>	TSCG as a Systemic Esperanto — GenericConcepts as a constructed neutral vocabulary enabling structural translation across disciplinary silos
H3	<code>LegoTechnic_Modularity.md</code>	Compositionality principle — M2 concepts as interlocking building blocks producing emergent structure through monoidal combination

1.17.2 F.2 Anti-Patterns & Safeguards

Document	Description
<code>OntologicalOverfitting.md</code>	The central anti-pattern: 1:1 map-territory correspondence as the ML overfitting analogue in ontology engineering. Defines the six-domain threshold as the primary guard
<code>_01_Narcissus_and_Icarus_as_Safeguards.md</code>	Two named epistemic failure modes: Narcissus (map absorbed into itself, Territory forgotten) and Icarus (vocabulary overreach, generality sacrificed). Operational safeguards built into the modeling methodology

1.17.3 F.3 Position Statements

Document	Description
<code>_00_TSCG_as_StereoscopicGlasses.md</code>	TSCG as a reading instrument — not proving, not criticizing, just seeing differently. The stereoscopic metaphor as epistemological stance
<code>CredibilityAccretion_Process.md</code>	How TSCG accumulates epistemic credibility through instance validation and community scrutiny — the falsificationist pathway to legitimacy

1.17.4 F.4 Extended Theoretical Foundations

Document	Description
<code>00_TSCG_Map_Territory_Theoretical_Foundation.md</code>	Map-Territory duality and observer relativity — theoretical foundation for future M3 extensions
<code>Archetypes_as_CrossCultural_Artifacts.md</code>	Jungian archetypes as proto-validated cross-cultural TSCG instances — empirical evidence for transdisciplinary invariants predating the framework by millennia
<code>TSCG_IntersubjectiveBenchmark_for_DefeasibleKnowledge_README.md</code>	Multisubjective Score Evaluation Protocol — methodology for quantifying and reducing inter-analyst variance in ASFID/REVOI scoring